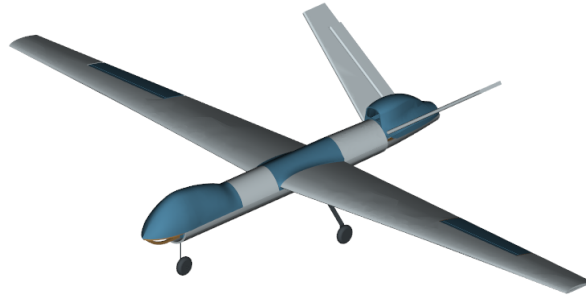


MQ9 / complete build guide

Manufacturing Team + Electronics Team



Start with your team's chapter. Use the contents and PDF bookmarks to jump to a task.

Cutting templates are separate in Manufacturing / Print_Templates. Print them at Actual Size / 100% and measure the calibration rectangle.

Contents

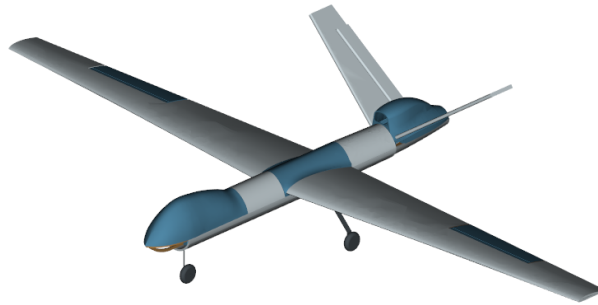
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Team coordination

Agree battery access and wire routes before closing the body. Record measurements and physical checks in Build_Record.csv. Complete the checks and acceptance chapter before flight planning.

01 / MQ9 airframe

Manufacturing and assembly guide / 2 metre RC model / Revision 1



For the **Manufacturing Team**: fabricate and assemble the body. The **Electronics Team** installs and configures electronics using its separate guide.

Start here

1. Read this guide and the part registers. Print the fit coupons first.
2. Cut plywood and make the printed/metal parts. Mark every part ID.
3. Dry-assemble the complete skeleton and removable wings.
4. Form and fit one representative foam panel, then finish the skins.
5. Deliver the body with access open for electronics, plus the completed inspection record.

Before flight: complete the fit, material, structural, propulsion and balance checks. Digital checks do not establish flight performance.

02 / Files, tools and stock

Folder	Use
01_Plywood	Individual CUT profiles, quantity register and revised positioned STEP parts.
02_Foam	Foam patterns, forming instructions and panel locations.
03_Print	Only selected aircraft print STLs, matching STEP and settings.
04_Hardware	Tube schedule, metal forming/drilling references, motor drawing and coupon.
05_Assembly	Revised assembly STEP and labelled reference illustrations.
Build Guide	Instructions, consolidated manual, part atlases and overviews.
Print_Templates	Actual-size A4 cutting and tooling templates. Keep these separate from the reading manual.

Machine work versus bench work

Laser / CNC: plywood CUT layer only.

Printer: selected aircraft STLs.

Metalwork: form and drill landing gear.

Knife: foam skins. Laser-cut XPS only with workshop approval for this material and process.

Bench work: dry-fit, bevel, form, glue and align parts; install fasteners and linkages; check moving clearances.

Stock and setup

Use aircraft plywood at the specified thickness, including 2.5 mm nose doublers. Skins use 5 mm FliteBoard Pro, 1000 x 600 mm. Use the specified carbon tubes; 5 mm solid rods are not substitutes.

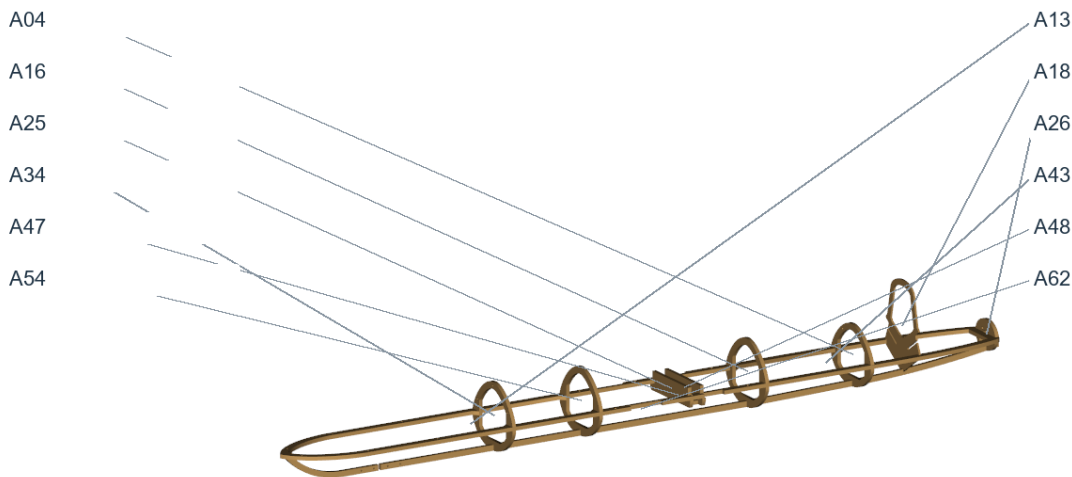
Sheet layouts use 1220 x 2440 mm stock. The shop may rearrange parts at 100% scale. Preserve quantities and grain along spars, spine and doublers.

03 / Parts and buying list

Item	Quantity / specification
Plywood	31 cut pieces from 22 profiles: 24 nominal 5 mm, 4 at 3 mm, 3 at 2.5 mm. Two outer ribs are 4.990 mm CAD thickness; finish 5 mm stock to fit.
Foam	5 mm FliteBoard Pro. Follow per-pattern quantities; moving tail controls use 2 layers each. Ten supplied sheets are starting stock, not ten installed sheets.
Printed aircraft parts	11 individual STLs. One fit coupon is additional and is not installed.
Carbon	Front 8/6 mm joiner 406 mm x 1; rear 8/6 mm 390 mm x 1; matched 10 mm OD/8.2 mm nominal ID sleeves 199.35 mm x 2.
Wheels and gear	3 lightweight 50.8 mm wheels, 3 mm bores; formed main gear and 3 mm spring-wire nose gear; steering arm and shaft collars.
Wing root retention	6 M2 bolts, start 20 mm long; 12 washers and 6 locking nuts. Verify final stack.
Nose bypass	2 M2x 40-45 inner bolts +2 M2x 16 outer bolts; 8 washers, 4 locking nuts. Select actual length from stack.
Main gear fastening	2 M3x 16 through-bolts, large washers, 2 locking nuts; 2 M3 axle bolts, nominal 25 mm plus spacers/nuts.
Motor fastening	4 M3 screws, matching washers; determine screw length from 5 mm ply and actual safe motor thread depth.
Nose bearing	4 M3 bolts through bearing/mount, washers and locking nuts; start 16-20 mm and verify stack/clearance.
Controls	5 servos: 2 ailerons, 2 tail, 1 nose steering. 4 control horns, 1 steering arm, 5 pushrods with 10 clevises/ball links; servo screws.
Hinge pins	2 smooth 3 mm rods; 285 mm initial length, trim after fitting retainers.
Adhesives / covering	Wood-compatible epoxy/wood adhesive; XPS-compatible foam adhesive; reinforced tape and paper seam straps. Qualify on offcuts.

Fastener lengths are starting sizes. Select final lengths from the actual washer, nut and part stack. Buy spare washers, locking nuts, cable ties and servo extensions.

04 / Fuselage skeleton



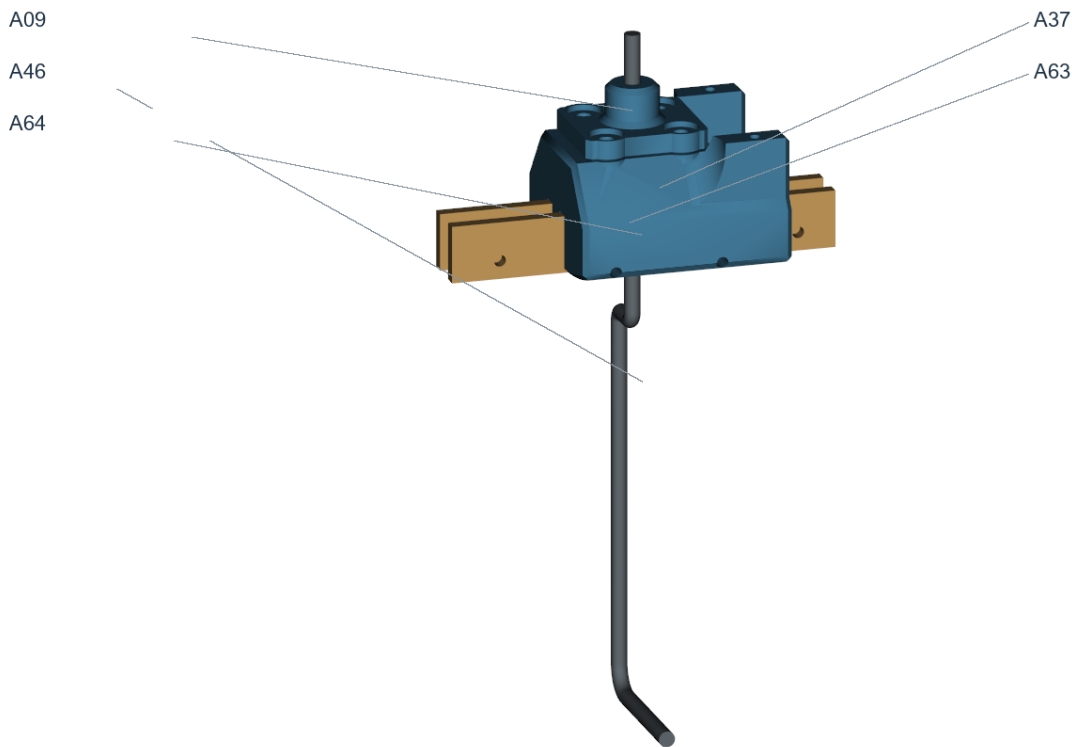
Assembly axes: X = span, Y = nose to tail, Z = up. Nose tip: Y = -439.65 mm. Use the revised assembly STEP for part positions.

1. Lay A62F and A62R on a flat jig along the centreline; first make the nose bypass on page 5. Do not glue them end-to-end across the shaft gap.
2. Fit forward longitudinal member A13 and aft member A43, then former A34 atY=-130, A54 atY=-5, A16 atY=233.65 and A04 atY=384.65. Slots determine final seating.
3. Fit the main-gear box A25/A48 with bottom plate A47. Fit tail bulkhead A18 and revised motor mount A26.
4. Check squareness from the assembly axes, equal left/right former extents, and that all tabs seat without forcing. Test fit the printed nose, centre and tail fairings before bonding.

Bond after dry-fit. Add modest fillets at gear, wing-root and motor joints. Keep electronics access open. Adjust the cutting process using a coupon; never scale a DXF to correct slot fit.

Spine: install A62F/R with the bolted side cheeks. There is no separate A40 piece.

05 / Nose gear bypass



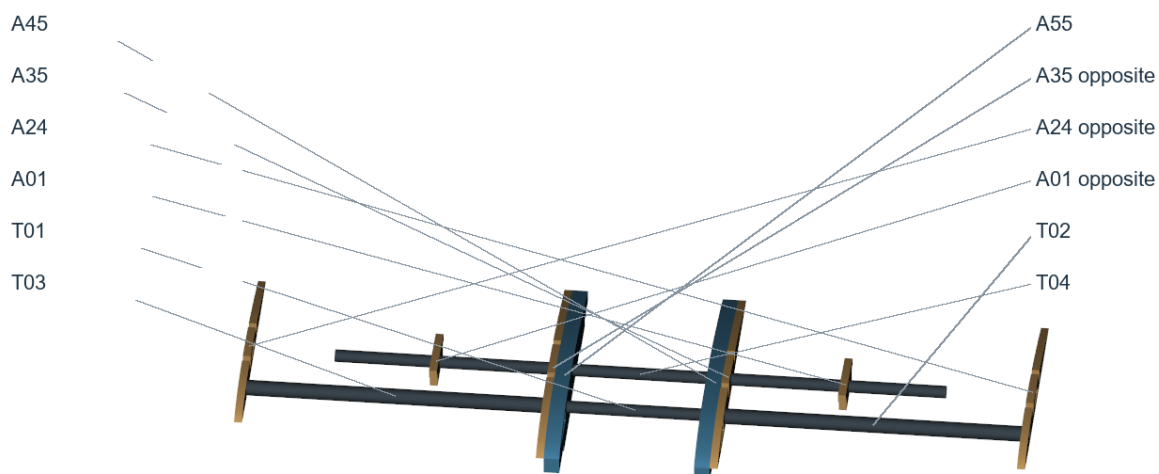
Nose joint: two 2.5 mm plywood cheeks, A63/A64, bridge the shaft gap. Each is 72 x 10.7 mm, with grain along its length. Fit them into the A37 pockets.

1. Place A62F/R collinear with the 3.4 mm shaft gap centred on $Y = -293.65$.
2. Fit one cheek on each side of the 5 mm spine. Dry-fit them into the revised A37 pockets.
3. Install four transverse M2 bolts at $Y = -320, -302, -282, -264$ mm, $Z = -37.964$ mm. Inner bolts pass through the mount; outer bolts clamp the wood stack. Use washers and locking nuts.
4. Fit A09 and the extended 3 mm wire. Rotate the shaft fully. Bond wood interfaces only after the assembly stays straight and turns freely; keep glue out of the shaft passage.
5. Tighten enough to seat the stack without crushing plywood/plastic. Recheck after cure.

Fit the 3 mm wire through A09 into its steering arm/collar. The supplied shaft extends to $Z = +1$ mm. Trim only after full clamp engagement. Retain mechanically; do not rely on glue.

Check: support the body and turn the wheel both ways. Check mount, cheeks and skin together for rubbing, cracks and loose bolts. Recheck under the intended gear deflection.

06 / Carbon joiners and retention



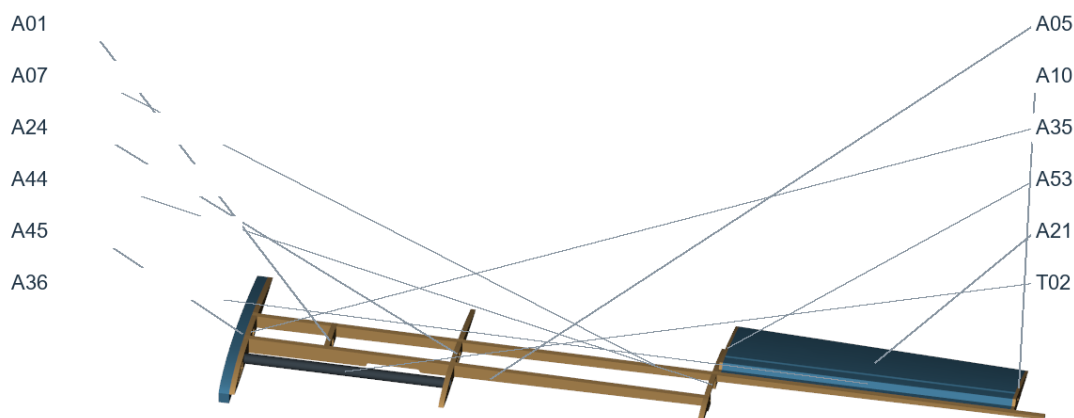
Part	Section / cut length	Position and purpose
T01	8 OD /6 ID; 406 mm	CentreX=-203..203. Front joiner axisY60.454, Z15.632.
T02/T03	10 OD /8.2 nominal ID; 199.35 mm each	RightX53..252.35; leftmirror. Bond sleeves into wing root/middle ribs.
T04	8 OD /6 ID; 390 mm	CentreX=-195..195. Rear axisY100.500, Z15.070.

Use a matched sliding tube pair. Nominal 8.2 mm sleeve ID gives 0.2 mm diametral clearance on the 8 mm joiner. Check actual stock. Do not force tubes or sand away structural wall thickness. Wood bores: 10.2 mm for sleeves; 8.2 mm for root/rear supports.

1. Centre both joiners and dry-fit both wings together.
2. Check about 150 mm front-joiner insertion per wing.
3. Bond sleeves only; keep glue off the sliding joiner.
4. Fit three M2 retainers per wing and mark tube insertion positions.
5. Check twist, spanwise retention and hinge clearance.

Use only the four tubes in the schedule. The 5 mm solid carbon rods are spare material.

07 / Wing frames and ailerons



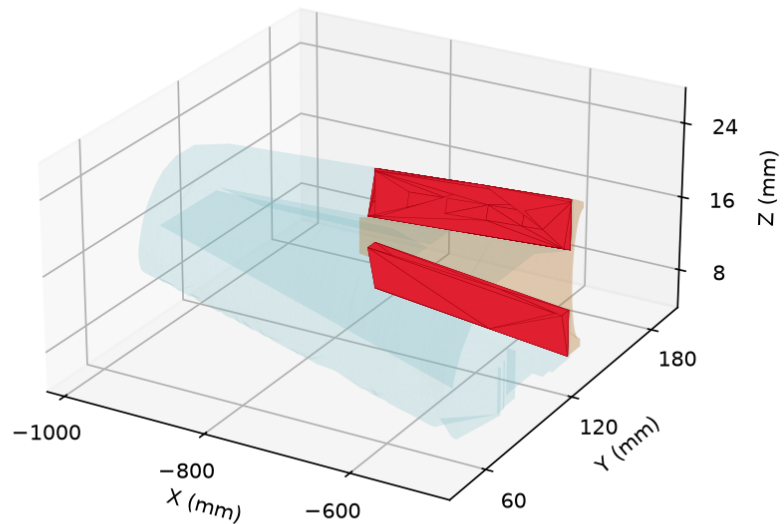
Build a right wing and its left-hand mirror. Follow register quantities and assembled STEP positions.

1. Fit inner rib A35, middle rib A24 and outer rib A44 to front spar A05 and rear spar A07. Fit rear joiner support A01 and the front sleeve. Match each rib slot, not a guessed spacing.
2. On a flat jig, check leading/trailing edges and equal incidence before bonding. Finish A24/A44 edges to the marked noncut sanding guide where needed; preserve revised bore centres.
3. Install the aileron end plates A53/A10, hinge rail A36 and aileron A21 on the right; use A23/A11 on the left. Dry-fit smooth 3 mm pins and retain them mechanically, leaving about 1 mm axial freedom.
4. Install each servo with its manufacturer screws and a serviceable mounting pad. Transfer actual servo mounting holes after confirming its size; the CAD servo is a reference envelope.
5. Fit the control horn and neutral pushrod before closing the skin. Check travel with no scraping, stiffness or exposed sharp pin end.

Check: ailerons rotate freely on retained pins. Keep glue out of hinges. Finish a tight bore to the actual pin; do not scale the STL.

08 / Foam skins and seams

Wing hinge inner core | red = CAD interference to relieve



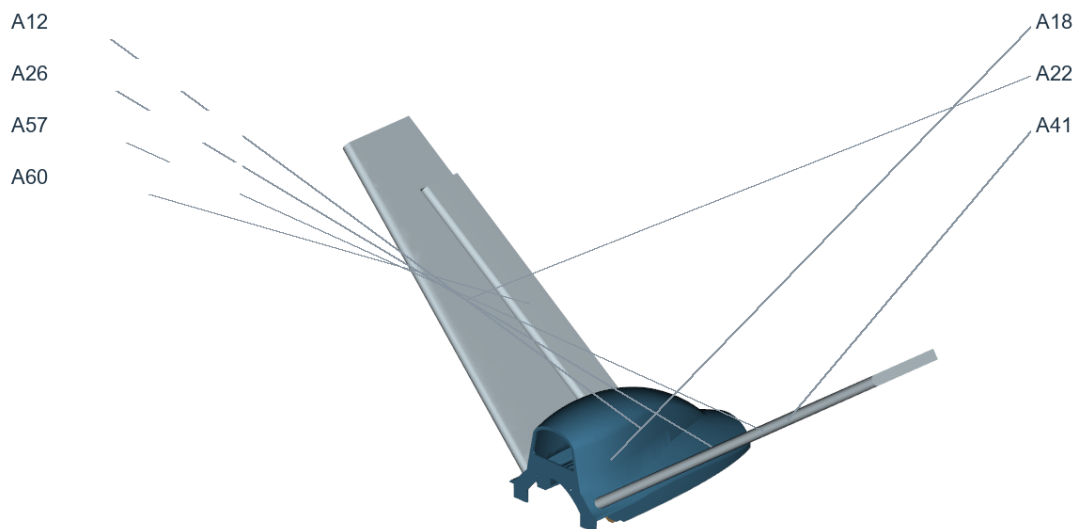
Blue: A27 foam. Tan: A23. Red: trim contact region. View cropped; axes not equal scale.

Use F-series patterns. Print at 100%, measure the calibration rectangle, and join matching marks with the indicated 10 mm overlap.

1. Mark each outside face and part ID. Knife-cut the finished skin edge. Moving-tail blanks have 5 mm sacrificial shaping stock. F23/F24 include a 1 mm PAPER-ONLY lower seam lip: remove its core and trim/fold it against the mating lip as shown in the foam guide. Other skin CUT lines have zero added lap/trim.
2. Peel inside paper only in the curved forming zones. Keep outside paper intact as the pattern reference. Gradually relieve/compress inside foam and dry-form over the real frame; do not stretch a short paper edge into place.
3. Seat the wing from its leading-edge region toward the open/trailing seam. Shave local inner foam for the printed hinge rail and rib contact patches. Do not cut away the outside paper to hide an inner interference.
4. Bevel inside seam edges to their actual meeting angle. Butt outside paper edges and add 15 mm internal paper straps, about 7.5 mm per side. Reinforce exposed seams with suitable light tape.
5. Dry-fit an entire wing and its moving surface before gluing progressively along the frame. Hold alignment while adhesive cures.

Check a forming coupon first. Peeled board loses stiffness and may warp. Hold both sides evenly while bonding. Use XPS-compatible adhesive and reinforce inner seams. Ordinary CA must not touch exposed XPS unless explicitly approved for it.

09 / Fuselage covering and tail



Dry-fit A59 nose, A06 centre fairing and A12 tail fairing. Keep a removable access panel for the battery, FC and wiring.

Follow the foam assembly map. Fit lower forward, middle and upper panels, then rear-underside bands in numbered order. Support butt seams inside. Relieve only inner foam where it contacts the spine or longitudinal members.

Trace installed gear before cutting exits. Start with 0.5 mm clearance around static legs, then check steering and deflection. Reinforce the paper edge. Gear loads must pass into the structure, not the foam.

Fixed tails: form skins with the removable collars; test A12 slot engagement.

Moving tails: laminate two 5 mm blanks. Transfer the finished outline, remove the 5 mm sacrificial perimeter and shape long edges to R5.

Tape-hinge on the indicated hinge line. Fit two servos, horns and pushrods with adjustment access. Check both tails for symmetry and free travel before glue cures. Keep straps and glue out of moving gaps.

10 / Printing and metalwork

Print group	Starting process settings, then qualify on a coupon
A06/A12/A59 shells	LW-PLA preferred for mass. If using available PLA+, start 0.2 mm layers, 2 perimeters and 3-5% infill; record slicer mass and weigh the finished part. Support internal overhangs without blocking sockets.
A11/A21 ailerons; A23/A36 rails	Start 3 perimeters, 10% infill; orient long axis on the bed with supports kept out of hinge bores. Use smooth pin sample and full travel check.
A09/A37/A45/A55 functional parts	Prefer a shop material/process suited to load and temperature; available PLA+ requires inspection/testing. Start 4 perimeters, 30% infill. Orient so tube/bolt loads do not simply split layer interfaces. Do not approve from infill percentage alone.

Print in millimetres at 100%. The shop selects orientation and supports. Some parts exceed 220 mm beds: use a larger printer. A smaller printer needs a designed joint, not an arbitrary split or rescale.

Check: clear every passage, measure dimensions, inspect layer bonds and reject cracked or warped brackets. Weigh printed parts for the Electronics Team.

Landing gear

Main gear A02: 12.7 mm wide x 3.175 mm metal strip. The DXF is a formed front view, not a flat blank. The shop selects bendable alloy/temper, bend radii and allowances while preserving attachment and axle positions.

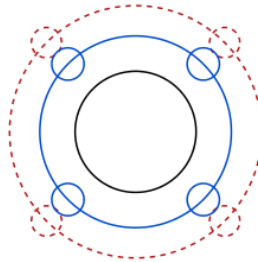
Fit A02 through its two 3.2 mm holes to A47 using M3 bolts, washers and locking nuts. Fit three 50.8 mm wheels. Form the 3 mm nose wire to its STEP reference. Check gear deflection at completed aircraft mass.

11 / Motor mount and team transfer

A26 motor mount - declared drilling correction

Source: 4x3mm on 25mm PCD. Revised: 4x3.2mm on 19mm PCD.

Physical Emax motor check required before drilling; drawing below at 2:1.



Red dashed: obsolete 25mm PCD - DO NOT DRILL

Blue: 19mm PCD - four 3.2mm holes

Central 12mm hole unchanged. Minimum radial wood web 1.9mm; use washers.

A26 motor mount: four 3.2 mm holes on a 19 mm pitch circle; adjacent hole centres 13.435 mm. Preserve the 12 mm centre opening.

Check the actual motor against the template before production. Use washers and measured screw engagement; screws must not reach windings. Keep wires clear of the motor bell.

Propulsion check required. Emax recommends 6-7-inch props. The 9-inch prop / 40 A ESC combination needs measured current, thrust and temperature checks. Keep the propeller off during assembly and calibration.

Leave these accessible

Keep access to battery straps, FC, ESC cooling, receiver, GPS, pitot tubing, wing connectors, horns, steering and gear fasteners. Fit wire pull strings before closing skins.

Transfer the body to the **Electronics Team** with recorded masses, fit notes and labelled loose hardware. Use the build record for outstanding checks.

12 / Checks before the body is signed off

Check	Builder records
Scale and coupons	100 x 20 mm rectangle measured; actual ply/foam thickness; kerf/slot result; printer coupon result.
Structure	All 31 wood pieces accounted for; no source A40 duplicate; bypass bolts installed; no split grain/cracks; aligned centreline.
Wing attachment	Both wings fit simultaneously; matched sleeve fit; centred tubes; six M2 retainers; no spanwise escape.
Controls	Two ailerons, two tail controls and nose steering free; hinge pins retained; horns secured; no glue in moving gaps.
Skins	Full surface dry-fit; seams meet without forced stretching; core relief/gear exits complete; outside paper sound; no wing twist.
Gear and motor	Positive gear fasteners; free wheels; free steering; motor pattern/screw engagement checked; ground clearance examined.
Mass and access	Total body and individual print masses recorded; battery/FC access open; wires can be serviced.
Outstanding evidence	Material forming, adhesive/print strength, gear loading, propulsion and flight checks must be recorded physically.

Balance reference for the Electronics Team

Provisional mark: 522 mm aft of the nose, or 22.2 mm aft of the front joiner. Leave battery travel available. This geometric quarter-MAC reference is not an approved flight CG range.

Balance in complete flight configuration. $CG\ movement = battery\ mass / total\ mass \times battery\ movement$. An experienced fixed-wing pilot must review stability, control authority and the final CG before flight.

Digital evidence

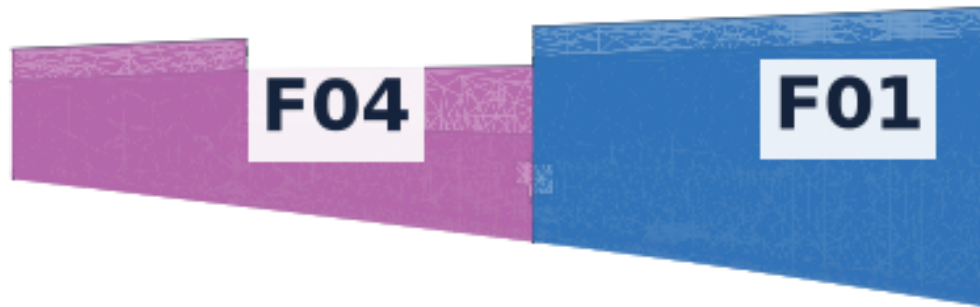
See the checks summary for recorded profile, mesh, foam and PDF-scale results. Record physical fit and load checks during the build.

Design basis: Onshape geometry, build transcript, Emax motor drawing and material specifications. CAD illustrations show part locations; colours distinguish materials.

Wings: match each skin to its side and span section

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

LEFT AIRCRAFT SIDE (-X)



RIGHT AIRCRAFT SIDE (+X)



 F04 Left outer wing A27

 F01 Left inner wing A52

 F02 Right inner wing A49

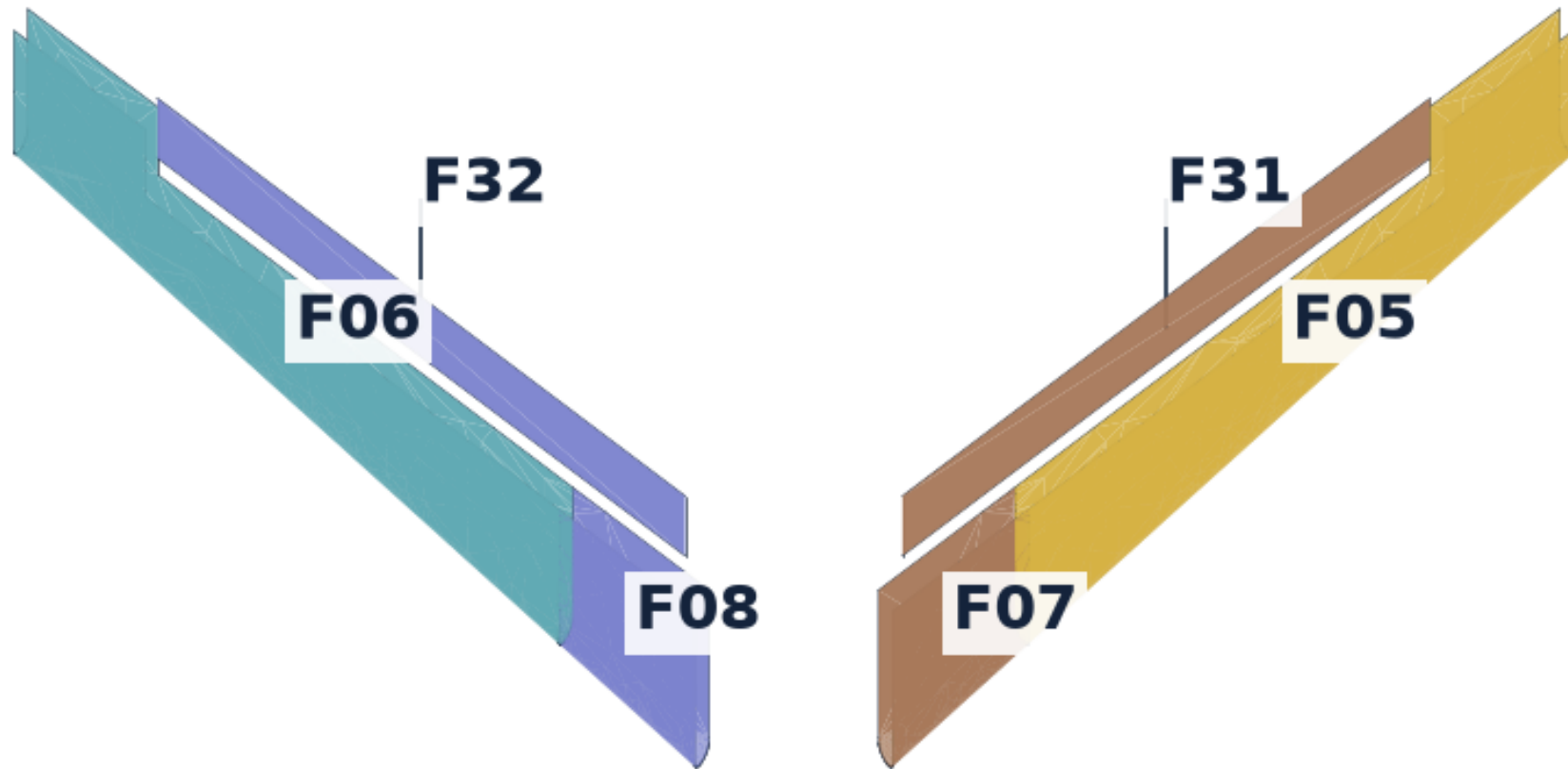
 F03 Right outer wing A38

Fit inner and outer sections to their ribs and spars before closing the trailing seams.

The narrow hinge-rail contact zones need inner-core relief; preserve the outer paper. See the forming guide.

V-tail: two fixed skins and one moving control on each side

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates



- F06 Left main skin A22
- F08 Left root collar A22
- F32 Left control A60: cut two layers

- F05 Right main skin A57
- F07 Right root collar A57
- F31 Right control A41: cut two layers

Each fixed-tail pair meets at the root collar seam (80% of the CAD span parameter from the tip).

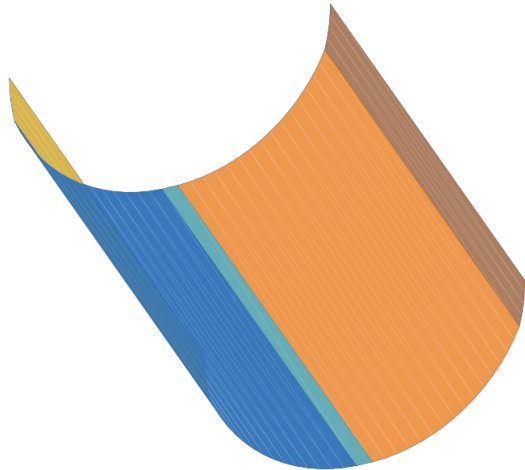
Control highlights show the CAD broad faces. Laminate two 5 mm blanks to 10 mm, then shape the R5 edges.

Use the numbered 1:1 DXFs / tiled PDF for cutting. Preserve the outside paper. See the forming guide.

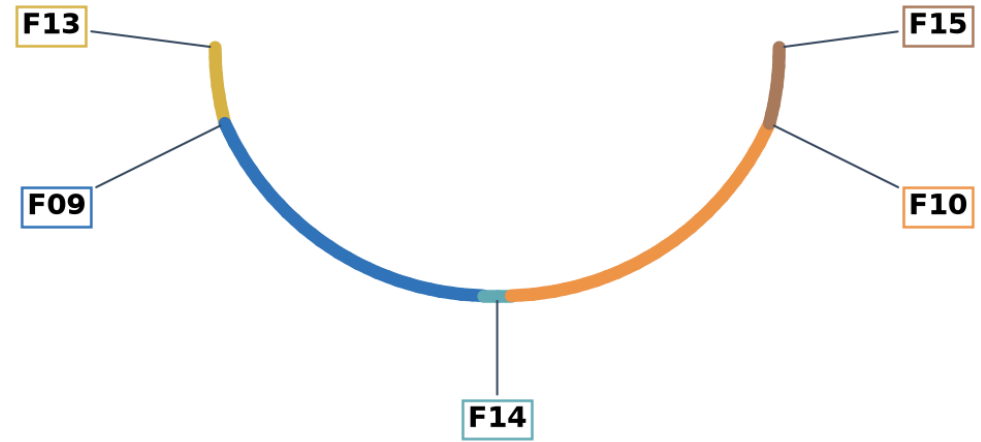
Lower fuselage: five distinct pieces around each section

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

FORWARD LOWER SHELL A56 | Y -335 to -130 mm

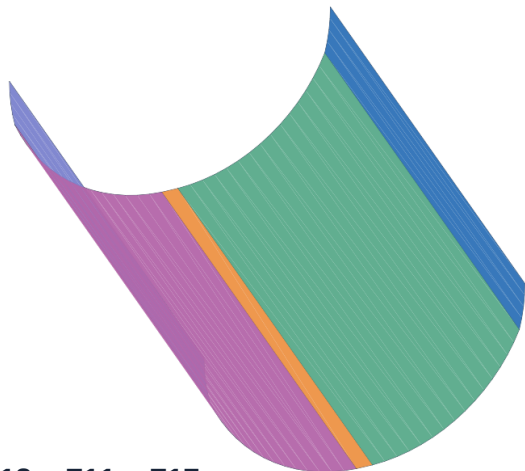


F13 > F09 > F14 > F10 > F15

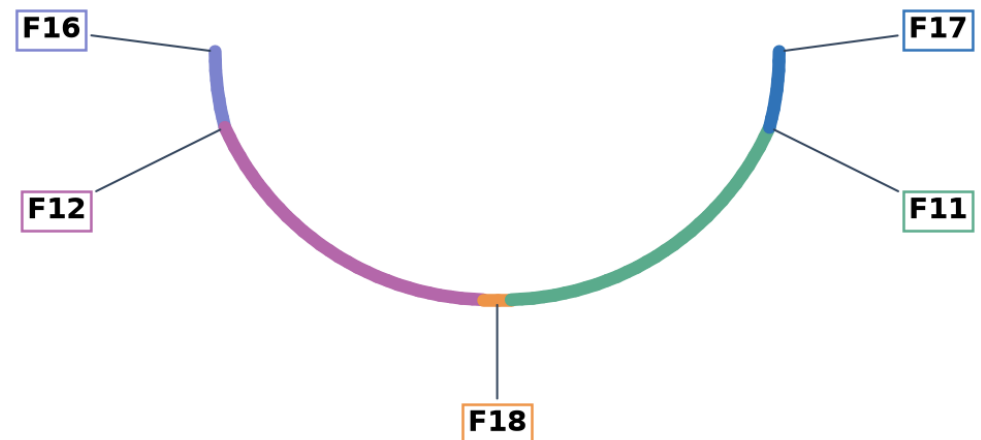


Cross-section: left to right around bottom

MIDDLE LOWER SHELL A17 | Y -130 to +238.65 mm



F16 > F12 > F18 > F11 > F17

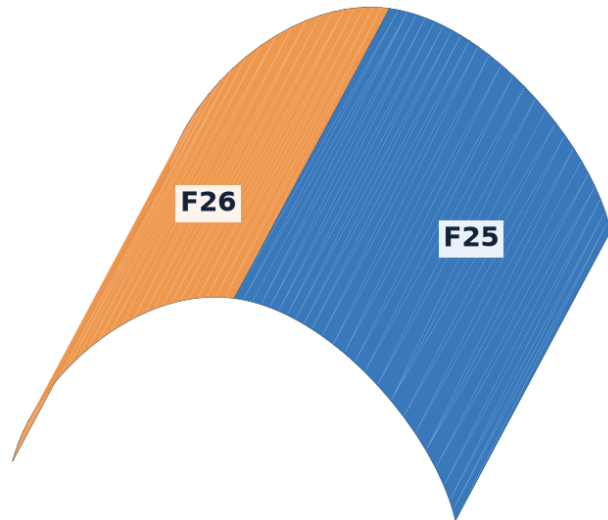


The narrow bottom strips are real parts; retain them.

Upper and aft-lower fuselage: match the coloured face pairs

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates

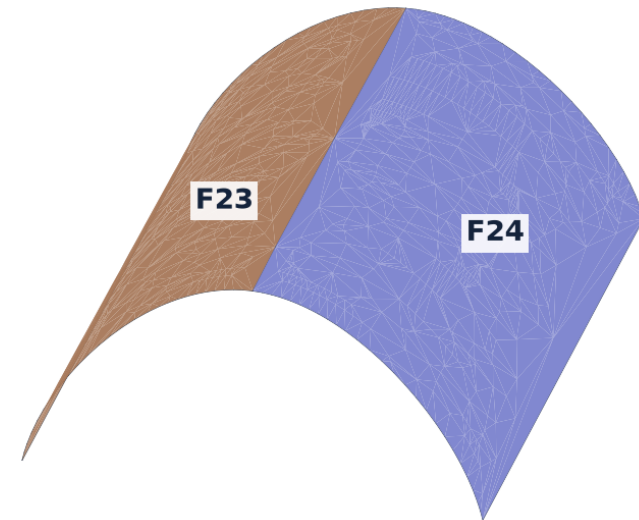
FORWARD UPPER A08 | Y -130 to 0



F26 left / F25 right; meet at the top.

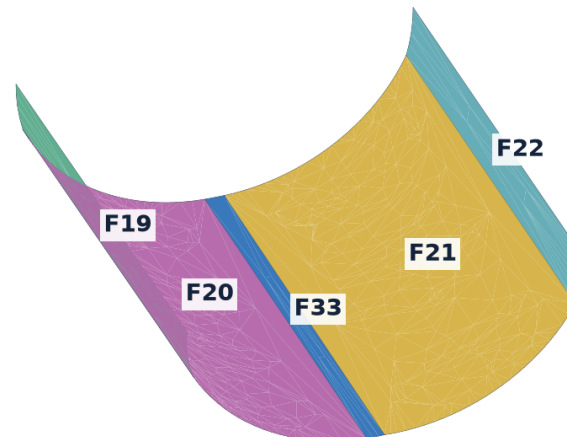
F23/F24 include a 1 mm paper-only lower-edge return: remove its foam, then trim the paper to fit.

AFT UPPER A15 | Y 233.65 to 389.65



F23 left / F24 right; meet at the top.

AFT LOWER A42 | Y 238.65 to 389.65 mm



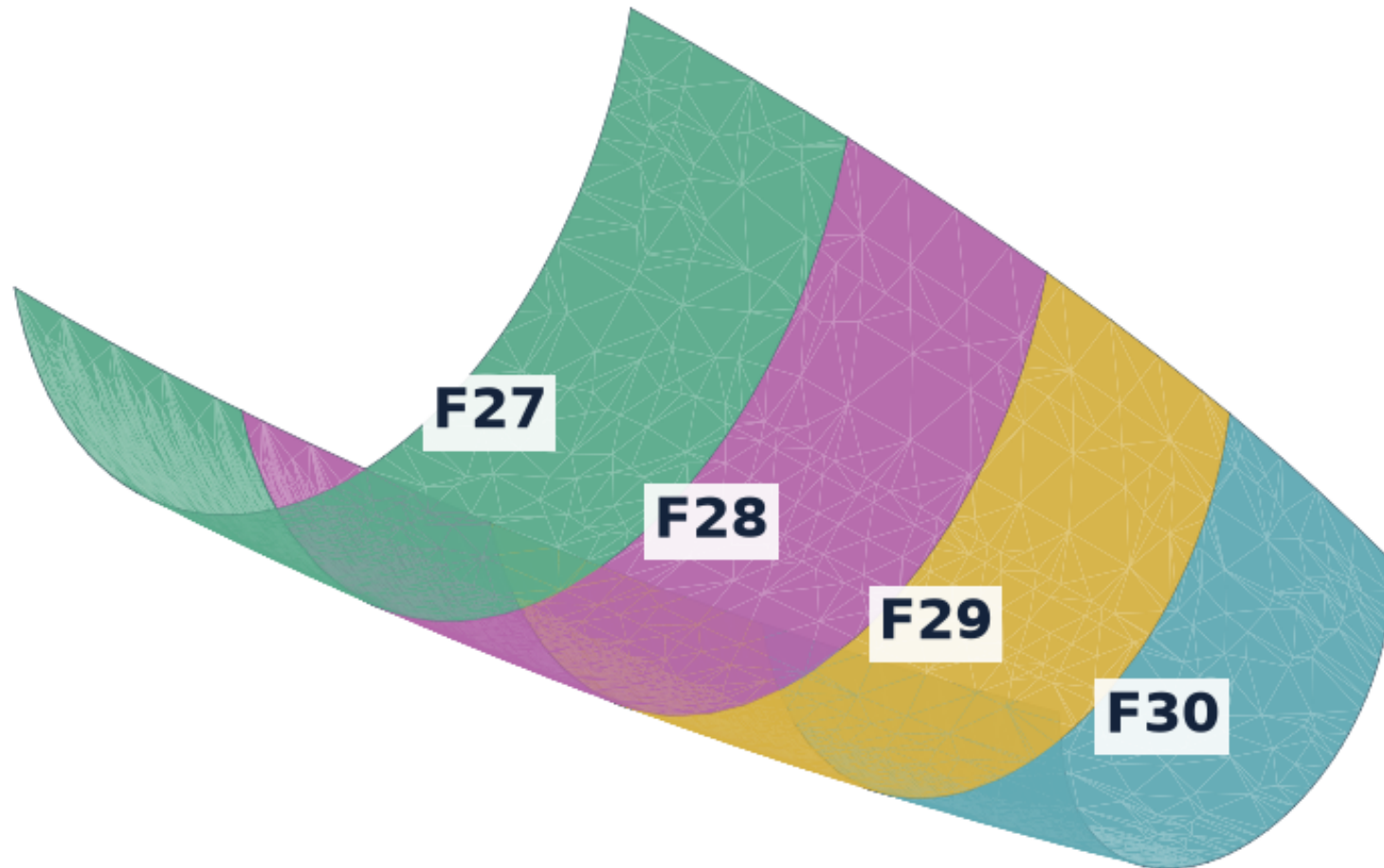
Around the underside: F19 left lip > F20 left curve > F33 bottom > F21 right curve > F22 right lip.

Fit the printed transitions between these shell groups before final skin bonding.

Use the numbered 1:1 DXFs / tiled PDF for cutting. Preserve the outside paper. See the forming guide.

Rear underside: assemble four bands from front to aft

MQ9 | 5 mm FliteBoard Pro | CAD surface assembly map | illustrations not cutting templates



 F27 1 - forward band

 F28 2

 F29 3

 F30 4 - aft band

A20 assembly order: F27 > F28 > F29 > F30. The Y coordinate increases toward the tail.

Butt the outside-paper seam lines. Add separate inside straps; dry-fit around A43 and A62 before gluing.

Use the numbered 1:1 DXFs / tiled PDF for cutting. Preserve the outside paper. See the forming guide.

Foam / prepare and cut

Stock

Use 5 mm FliteBoard Pro paper-faced XPS, 1000 x 600 mm. Measure actual thickness. Do not resize patterns for another thickness.

1 / Print and identify

Print at Actual Size / 100%. Measure the 100 x 20 mm rectangle. Join only matching S-number posters, using row/column labels, paired marks and 10 mm overlap. Follow the index for whole-page patterns.

2 / Make a forming coupon

Keep the outside paper. Peel inside paper in curved areas and gradually relieve or compress the core. Test the tightest intended curve and the adhesive on scrap before cutting the full set.

3 / Cut patterns

Knife-cut CUT lines and transfer REFERENCE marks and panel IDs. Mark OUTSIDE before peeling. DXFs are millimetres with no tool compensation. Use a laser only if the workshop approves this exact XPS and extraction process.

Skin seams

CUT is the finished outside-paper edge: no extra lap or trim, except F23/F24. Those two have a 1 mm paper-only lower lip: remove its foam, keep the paper and trim/fold it against the adjoining lip.

Moving-tail blanks

Cut F31/F32 twice each. Each blank includes 5 mm sacrificial stock around the reference outline. Laminate two 5 mm layers, then trim to the finished outline and shape long edges to R5.

Adhesive

Use XPS-compatible adhesive verified on scrap. Ordinary CA must not contact exposed XPS unless its manufacturer approves that use.

Foam / wings and seams

1 / Dry-fit the frame

Align ribs, spars and both removable wings. Check incidence and symmetry before covering. Match F-series IDs to the assembly map; A/B does not automatically mean left/right.

2 / Form from the leading edge

Keep the outside paper continuous. Peel inside paper only in curved forming zones. Press progressively over the matching frame. Do not stretch a short edge to force a seam.

3 / Relieve the inner core

Shave inner foam where ribs or hinge rails contact it. Preserve outside paper. Fit the whole wing and moving surface together before bonding.

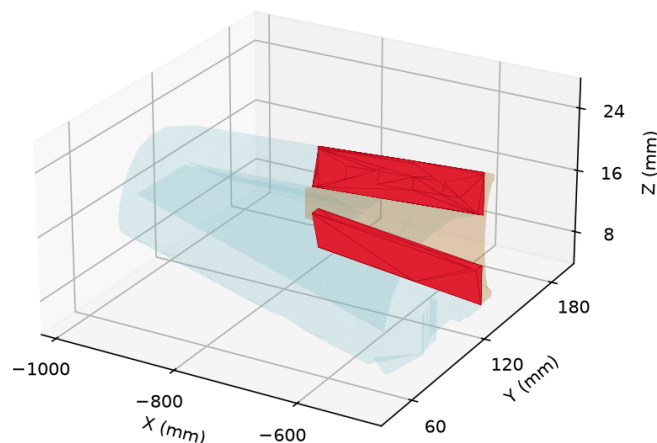
4 / Close seams

Bevel inner foam to the actual meeting angle; there is no universal 45-degree bevel. Butt outside-paper edges. Add separate 15 mm internal paper straps, about 7.5 mm each side. Keep straps out of hinges.

5 / Bond and restrain

Glue progressively with the wing held straight. Peeled board loses stiffness and can warp. Reinforce seams and relief borders without adding material that recreates an interference.

Wing hinge inner core | red = CAD interference to relieve



Blue: A27 foam. Tan: A23. Red: trim contact region. View cropped; axes not equal scale.

Foam / fuselage and tail

Fuselage order

Fit lower forward, middle and upper panels on the aligned frame. Fit rear-underside bands F27-F30 in numbered order, supporting every inner butt seam. The A42 underside sequence is F19, F20, F33, F21, F22; do not omit F33.

Gear openings

Trace the actual installed gear. Start with 0.5 mm static clearance, then check steering and deflection. Reinforce paper at the opening; foam must not carry gear loads.

Fixed-tail forming tools

Use the removable split collars on a straight jig: T03 at 6.84 mm, T02 at 171 mm and T01 at 335.16 mm along the 342 mm span. Make tools from stiff card or 3 mm scrap plywood; remove both halves after cure.

Fixed-tail skins

Fit the paired skins and root-collar seam. Relieve inner foam so walls meet while preserving the outside shape. Do not add internal ribs into the filled section. Check fit in A12 and symmetry before bonding.

Moving tails

Laminate and shape the two controls. Tape-hinge along the indicated line, fit horns and linkages, and check full travel. Keep glue and seam straps out of moving gaps.

Before closing

Check battery/FC access, cables, joiners, control movement and symmetry. Weigh completed skins and body. Record the physical fit and forming results.

Foam / local fit checks

Wing hinge area

Relieve inner foam around hinge rails. Never cut through outside paper to hide a contact.

Lower centre skin A17

Mark A45/A55 bracket contacts inside and relieve only the foam. Do not sand structural brackets or plywood to fit the skin.

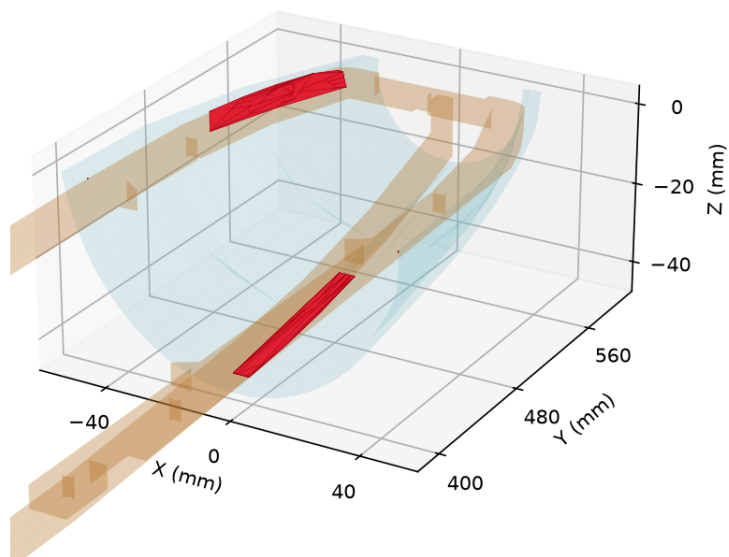
Rear underside A20

Fit A43 and the spine before F27-F30. Transfer contacts into the inner foam. Keep seam support continuous across reliefs.

Final check

Repeat dry-fit with all neighbouring parts installed together. Reinforce relieved-zone edges and seams. Digital pattern development does not establish the stiffness of peeled or relieved foam.

Rear underside inner core | red = CAD interference to relieve



Blue: A20 foam. Tan: A43, A62. Red: trim contact region. View cropped; axes not equal scale.

Electronics build guide

Matek F405-WING V2 / ArduPlane / FlySky / five servos with nose steering

For the **Electronics Team**. Follow the steps in order and record actual readings in the build record.

Remove the propeller for wiring, firmware, calibration, servo and failsafe work. Refit it only for a separate controlled propulsion test after bench checks pass.

Equipment	Build assumption
Flight controller	Matek F405-WING V2; not V1, WTE or TE
Navigation	Matek M9 GPS; exact SKU and compass presence unconfirmed
Controls	Two aileron + two V-tail + one nose-steering servo for supplied gear; 9 g metal gear, model unknown
Power	GenX 4S 5200 mAh LiPo; ReadytoSky 40 A 2-4S ESC advertised as Opto with 5 V / 3 A BEC
Propulsion	Emax ECO II 2807 1300 KV + Gemfan 9045 CL three-blade; combination unqualified
Radio / airspeed	FlySky FS-i6X + FS-IA10B; Matek airspeed SKU unconfirmed

Work in this order

Identify parts -> build and meter harnesses -> configure firmware -> connect radio and sensors -> centre and test servos -> test failsafes -> install and balance -> qualify propulsion. Save a parameter file after each successful stage.

Five servos: two ailerons, two tail controls and one nose steering. Establish control throws, flight CG, speeds, gains and return altitude from the completed aircraft and measured tests.

Resolve these labels first

Connector shape and wire colour are not a pinout

Unknown	Safe branch before connection
M9 GPS SKU	Read PCB and module labels and obtain that Matek manual. UART module: TX to FC RX3, RX to TX3; match specified supply and ground. Add SDA/SCL only if an I2C compass exists. CAN-only module: stop; this FC has no native CAN. Never infer pin order from another M9.
Airspeed SKU	ASPD-4525: I2C, ARSPD_TYPE=1. ASPD-DLVR: labelled I2C connector, ARSPD_TYPE=9; its CAN connector is different. Verified I2C branch: DA2/SDA + CL2/SCL, ARSPD_BUS=1. Other SKU: use its manual and leave disconnected until identified. [1,6,7]
ESC signal plug	Determine whether red is BEC output, absent, or required optocoupler supply. If BEC output, depin and insulate individually; retain signal and signal ground. If input-side power is required, follow the exact manual. Do not guess from Opto in a listing.
9 g servo model	Confirm voltage, pulse range, current and torque. Use 5 V only if supported. Do not select 6/7.2 V from size or metal gears alone. Five servos cover the supplied controls and steering; quantity is not torque approval.
Battery / connectors	Record mass, dimensions, connector polarity and charge limits. Meter every adapter. Never connect the 4S main lead to receiver, GPS, servo rail or 5 V pad.

Make a connector record

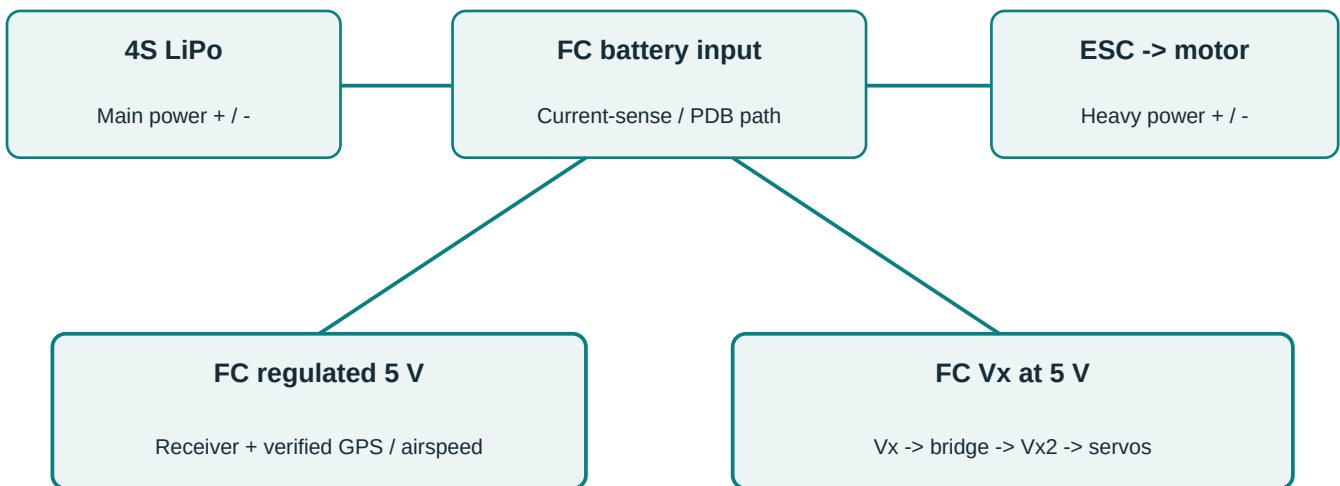
Draw each plug from a stated face. Number its cavities and label signals. With power removed, continuity-check both ends and label the harness.

Pad names below identify functions. Verify connector cavity numbers from the exact hardware manual before soldering.

Power architecture

One positive source per rail; common signal ground

Route battery and ESC power through the FC power-distribution pads. Match the V2 board layout [1].



ESC signal + ground -> FC S1 + G

ESC BEC output (if present) -> insulated; never joined to FC rails

Build and meter

1. Fit strain-relieved battery and ESC power leads to the documented pads. Keep high-current loops short. Select connectors and wire for measured load; insulate exposed joints.
2. Keep electronics 5 V and servo Vx positives separate. S5-S9 use Vx2: bridge Vx to Vx2 for this internal-BEC plan. Check before plugging in servos. [1]
3. If tests require an external BEC: remove the bridge, verify isolation, power Vx2 from that BEC and move all servo positives to that rail. Retain grounds; never join two BEC positives.
4. With peripherals unplugged, check polarity and hard shorts. First power through a suitable current limiter, then meter 5 V and Vx/Vx2 at far connectors. A limiter may brown out under servo load; it does not qualify flight power.

Wire the signal harness

FC outputs and transmitter channels are independent

FC connection	Destination / setting
S1 + G	ESC PWM signal + ground. SERVO1_FUNCTION=70 (Throttle). Isolate BEC output lead if present.
S2	Unused, SERVO2_FUNCTION=0; leave empty.
S3 + Vx + G	Left aileron. SERVO3_FUNCTION=4 (Aileron).
S4 + Vx + G	Right aileron. SERVO4_FUNCTION=4 (Aileron).
S5 + Vx2 + G	Left tail. Initially SERVO5_FUNCTION=79 (V-tail left). Validate page 8; physical motion governs.
S6 + Vx2 + G	Right tail. Initially SERVO6_FUNCTION=80 (V-tail right). Validate page 8.
S7 + Vx2 + G	Nose steering A39: SERVO7_FUNCTION=26 (GroundSteering). Omit only with a deliberately fixed nosewheel.
S8-S9 / LED	Unused; disconnect. Check SERVO8-10_FUNCTION for unwanted assignments.
RX2 + 5 V + G	FS-iA10B iBUS servo-output signal + supply + ground. Not SENS telemetry port. BRD_ALT_CONFIG=0.
RX3 / TX3 / supply / G	Verified UART GPS: module TX -> RX3, RX -> TX3; supply per exact SKU. SERIAL3_PROTOCOL=5 (GPS).
DA2 / CL2 / supply / G	Verified I2C compass / airspeed: SDA -> DA2, SCL -> CL2. Match pinout and supply. Check shared-bus address conflicts.

Custom output map: write and read back every assignment before connecting the ESC. S1 is throttle only after configuration. [2-4]

Start with supported 50 Hz PWM. Leave DShot disabled for this unidentified ESC. Check shared timer groups before changing rates. [2]

Firmware and first connection

Mission Planner on Windows; menu names can change

Install the matching target

1. Confirm **F405-WING V2**. Connect a data USB cable with battery, ESC and servos disconnected.
2. Install stable **ArduPlane / MatekF405-Wing**, version 4.4 or newer. Do not select Copter, INAV, F405-STD or F405-TE. [1]
3. With an ArduPilot bootloader, load the matching .apj. Otherwise follow the board DFU first-install procedure using firmware with its bootloader. [2]
4. Record target and version. Save initial parameters and inspect sensor messages. Insert a supported logging microSD with power off.

Configure before powering outputs

Configure: write page 4 settings in Full Parameter Tree/List. Reboot and read back. Use Mandatory Hardware for sensors, radio and modes; Servo Output for servos.

Do not bulk-import an internet parameter file. Keep arming checks enabled. Resolve each failed check instead of setting `ARMING_CHECK=0`. Save the parameters actually read from this board.

Recovery route

Recovery: disconnect propulsion power. Reconnect USB and restore the last known parameter file. Repeat sensor, output and failsafe checks after firmware changes.

Nose steering: centre A39 with `SERVO7_TRIM`; set safe endpoints and reversal so right rudder steers right. Push-test with motor off. Set `GROUND_STEER_ALT` deliberately. Stabilized steering needs healthy heading and `STEER2SRV_P` based on measured turning diameter. [14]

Radio input and modes

Plain controls from FS-i6X; mixing inside ArduPlane

Create an airplane model; select AFHDS 2A and bind the receiver. Disable transmitter V-tail/elevon mixes and centre trims. Use the receiver iBUS servo port with regulated 5 V. [5]

Input	Intended function
CH1	Roll; RCMAP_ROLL=1
CH2	Pitch; RCMAP_PITCH=2
CH3	Throttle; RCMAP_THROTTLE=3
CH4	Yaw; RCMAP_YAW=4
CH5	Three-position mode switch; FLTMODE_CH=5. Measure its actual detent values.
CH6 optional	Dedicated arm/disarm only after verifying the installed firmware option and behaviour; never share with mode control.

Calibrate

Calibrate inputs: power RX from verified FC 5 V. Check each stick moves the intended bar. Sweep sticks and switches; finish at low throttle with other sticks centred. Save. [8]

Input directions: right roll, pitch up, more throttle and right yaw must increase their normalized inputs. Correct RCn_REVERSED first; adjust surface direction later with SERVOx_REVERSED. Recalibrate after endpoint changes. [8]

Modes: initially set all six slots MANUAL. Assign the middle detent to FBWA. Keep the third MANUAL until navigation checks pass, then deliberately assign RTL. Check HUD mode at every detent and transition.

Receiver telemetry is not automatically flight-battery voltage. Keep antenna active ends clear of carbon and power wiring; range-test after final installation.

Sensors and battery readings

Calibrate in final orientation; inspect the actual response

Attitude, heading and position

Attitude: secure FC and set AHRS_ORIENTATION. Calibrate accelerometers in six positions, then level against the aircraft datum. Verify HUD roll/pitch follow the aircraft.

Compass: the FC has no onboard compass. Identify any external one, set orientation and calibrate away from metal and power wires. Verify heading in several known directions after installation. [2]

GPS: use SERIAL3_PROTOCOL=5 and the appropriate GPS type. Check outdoors for plausible position, fix and time. GPS-only hardware does not provide magnetic heading.

Battery monitor - V2 values

Parameter	Initial value
BATT_MONITOR / voltage / current pins	4 / BATT_VOLT_PIN=10 / BATT_CURR_PIN=11
BATT_VOLT_MULT / BATT_AMP_PERVLT	11.0 / 66.7; older-board 31.7 is wrong for V2
BATT_CAPACITY	5200 mAh nominal; validate usable capacity separately

Calibrate against a meter:

New voltage multiplier = old multiplier x meter voltage / reported voltage.

New current scale = old scale x reference current / reported current.

Use a steady meaningful current, not near zero. Check idle offset and consumed capacity. [1,4]

Set low/critical thresholds, delays, reserves and actions from pack specifications and measured loaded behaviour. Record them; the C-rating does not prove endurance.

Servos and V-tail direction

Left/right viewed from behind the aircraft, facing forward

Centre servos electrically before fitting horns. Align linkages mechanically; use SERVOx_TRIM for small corrections. Set conservative endpoints and check every hinge through full travel.

Test	Required response
MANUAL: roll right	Right aileron trailing edge up, left down; reverse for roll left.
MANUAL: pitch up	Both tail trailing edges move in upward pitch-producing sense; reverse for pitch down.
MANUAL: yaw right	Both tail trailing edges move toward aircraft right; reverse for yaw left.
FBWA, centred sticks: tilt aircraft right	Left aileron up, right down: correction opposes tilt.
FBWA: raise nose	Tail command produces nose-down correction; lowering nose produces nose-up correction.

V-tail correction: wrong in both pitch and yaw → reverse output. Wrong in only one axis → exchange functions 79/80 and repeat both tests. Physical motion governs the final mapping. [9]

Roll-to-rudder testing requires nonzero KFF_RDDRMIX; restore its intended value afterward. A stationary yaw test does not prove heading hold. [9]

Combined commands and supply loading

Test all four full pitch/yaw combinations. Check binding, buckling, buzzing and saturation. Adjust mechanics, endpoints and MIXING_GAIN; repeat MANUAL and FBWA tests. [9]

Load-test all servos together without prolonged stall. Capture voltage dips, current peaks, temperature and resets. A multimeter may miss brief dips. Limits: Vx 5 A continuous / 6 A peak; electronics 5 V rail 2 A. Verify margin. [1]

Airspeed and propulsion

Sensor response is not calibration; motor spin is not thrust

Airspeed branch

Identify sensor and set ARSPD_TYPE / ARSPD_BUS. ASPD-4525: type 1. DLVR-L10D I2C: type 9. Never connect its CAN socket to I2C. Reboot and check health. [6,7]

Put pitot in clean airflow, clear of prop wash. Match dynamic/static ports; avoid leaks and kinks. Let temperature settle. Shield from wind without sealing ports; run Preflight Calibration for zero. [10]

Apply gentle airflow: reading should rise and return near zero. Avoid pressure, liquid or blowing into tubing. Start ARSPD_USE=0; some Plane versions still use a healthy sensor for speed scaling. [10,11]

If readings are suspect, set ARSPD_TYPE=0 and resolve them. Ratio calibration needs a safely flying aircraft and log review. Bench airflow does not establish true airspeed or stall speed.

ESC and motor

Motor mount: four M3 holes on 19 mm PCD; adjacent spacing 13.435 mm. Check actual motor fit and safe screw engagement. Emax lists 6-7-inch props; test the proposed 9-inch combination before use. [15]

Confirm ESC PWM range and calibration method. FC/RX must stay powered while ESC power switches separately. Use a documented isolated setup or servo tester; do not backfeed power rails. [12]

If the ESC manual specifies high/low calibration: remove prop, switch ESC off, command maximum, power ESC, wait for its indication, then command minimum. Verify acceptance. Do not use multicopter calibration for Plane. [12]

Check stopped motor when disarmed and at minimum. Restrain the airframe for a brief low-speed rotation check. Power off before swapping two phase wires to reverse. Confirm pusher prop orientation produces forward thrust.

Failsafes: prove the trigger

Prop removed; disconnect motor power for configuration tests

With prop removed and telemetry recording, switch the transmitter off. Observe RC inputs, failsafe message, mode and recovery when the link returns.

Radio loss

Set `THR_FAILSAFE=1`. Check whether RX stops frames, flags loss or holds/presets channels. Held normal channels may hide link loss. Configure a detectable condition and repeat. [13]

For low-throttle detection, put `THR_FS_VALUE` between measured loss and valid-low throttle. Example only: loss 900, valid low 1000, threshold 950 us. Require clear separation.

Candidate policy: short loss `CIRCLE`, `FS_SHORT_ACTN=1`; sustained loss `RTL`, `FS_LONG_ACTN=1`. Verify firmware labels and both timeouts. Set valid home, return height, terrain clearance and power reserve before flight. [13]

Battery and navigation failures

Record low/critical voltage, capacity, delay and actions. Test with a suitable variable supply or temporary thresholds, then restore and read back flight values. Never deeply discharge the LiPo for a test.

A ground `RTL` mode change does not prove a safe return. Test navigation-failure policy in simulation, then controlled evaluation with an experienced pilot. Do not unplug live sensors near a running prop.

Record trigger, delay, action and recovery. Repeat from every intended flight mode. If a failsafe cannot be observed or explained, keep the flight gate closed.

Installation and release record

Complete on the actual aircraft

Restrain the adjustable battery. Separate power wiring from GPS, compass and receiver. Protect wires at edges, strain-relieve plugs and label wing/tail connectors. Keep USB, SD, rail bridge and battery access open.

Provisional setup balance station

Provisional balance mark: 522 mm aft of nose, or 22.2 mm aft of front joiner. This is geometric quarter-MAC, not an approved flight CG range. Check complete mass, control authority and final CG with an experienced fixed-wing pilot.

xCG = sum(mass x position) / total mass

Battery adjustment: battery shift = total mass / battery mass x desired CG shift.

Use one nose datum, aft positive. Include wiring, glue and finish. Mark repeatable pack positions; verify assembled balance and lateral balance.

Record	Measured / approved value
Firmware / saved parameters	_____
5 V minimum / Vx minimum / servo peak current	_____
Voltage calibration / current reference	_____
RC loss trigger / short / long / recovery	_____
Battery low / critical / actions	_____
Total mass / battery mass / actual CG from nose	_____
Desired CG shift / required battery movement	_____
Control throws / propulsion test / flight CG review	_____

Before flight: verify connectors, loaded supplies, control directions, range/failsafes, sensors, structure, CG and propulsion. Measure current, thrust, voltage and temperature on a proper restrained test fixture.

Manuals and reference links

Primary sources checked 19 September 2026

[1] Matek V2 specifications, layout and mapping

<https://www.mateksys.com/?portfolio=f405-wing-v2>

[2] ArduPilot F405-Wing (older details need V2 cross-check)

<https://ardupilot.org/plane/docs/common-matekf405-wing.html>

[3] ArduPilot Choosing Servo Functions

<https://ardupilot.org/plane/docs/servo-functions.html>

[4] ArduPilot board hardware definition (master)

https://github.com/ArduPilot/ardupilot/blob/master/libraries/AP_HAL_ChibiOS/hwdef/MatekF405-Wing/hwdef.dat

[5] FlySky FS-iA10B specification

<https://www.flysky-cn.com/ia10b-canshu>

[6] Matek ASPD-4525

<https://www.mateksys.com/?portfolio=aspd-4525>

[7] Matek ASPD-DLVR

<https://www.mateksys.com/?portfolio=aspd-dlvr>

[8] ArduPilot RC input mapping and reversal

<https://ardupilot.org/plane/docs/rc-throw-trim.html>

[9] ArduPilot V-tail / inverted V-tail setup

<https://ardupilot.org/plane/docs/guide-vtail-plane.html>

[10] ArduPilot Airspeed Calibration

<https://ardupilot.org/plane/docs/calibrating-an-airspeed-sensor.html>

[11] ArduPilot Airspeed Parameters Setup

<https://ardupilot.org/plane/docs/airspeed-parameters-setup.html>

[12] ArduPilot ESC Calibration

<https://ardupilot.org/plane/docs/common-esc-calibration.html>

[13] ArduPilot Plane Failsafe Function

<https://ardupilot.org/plane/docs/apms-failsafe-function.html>

[14] ArduPilot Ground Steering

<https://ardupilot.org/plane/docs/tuning-ground-steering-for-a-plane.html>

[15] Emax ECO II 2807 official drawing in product description

<https://emaxmodel.com/products/emax-eco-ii-series-2807-1300kv-1700kv-1500kv-brushless-motor-for-rc-drone-fpv-racing>

Keep the exact ESC, servo, GPS and airspeed manuals with the build record. Complete the measured bench and flight checks on the actual aircraft.

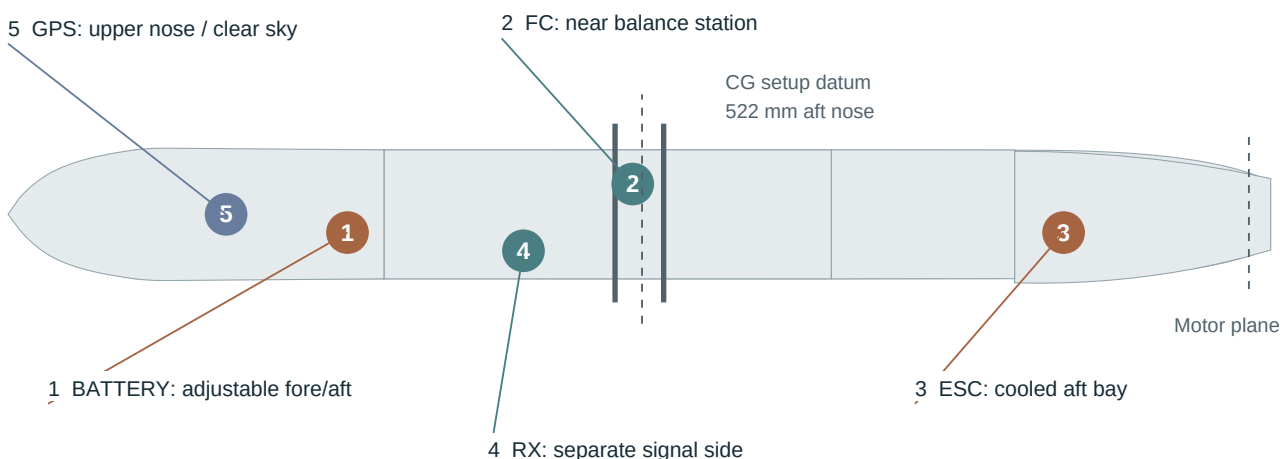
Where the electronics go

Proposed zones on the actual CAD body - not mounting-hole templates

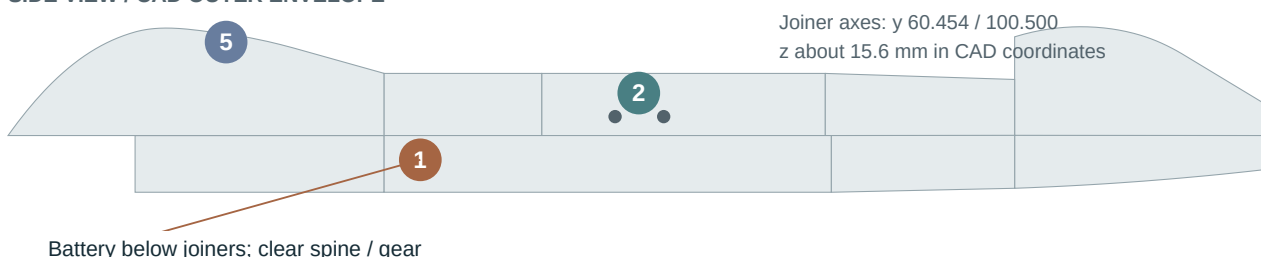
With the structure open, trial-fit the battery and equipment in these zones. Check actual battery size and cable exit before making its tray or hatch.

TOP VIEW / NOSE LEFT

PUSHER AFT



SIDE VIEW / CAD OUTER ENVELOPE

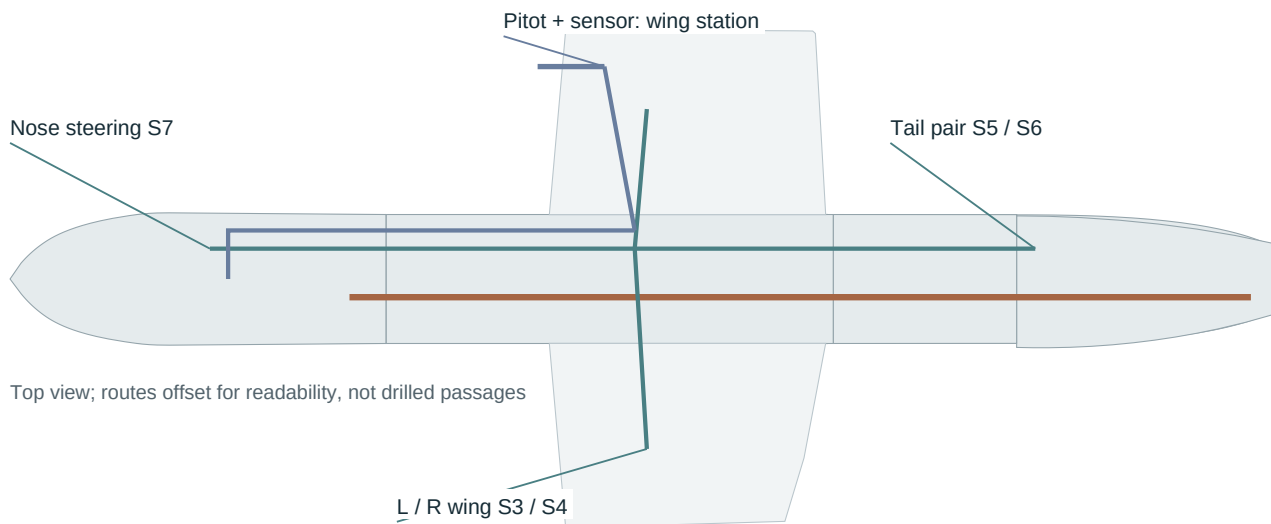


Zone	Mounting decision
1 Battery	Adjustable tray below the wing tubes, to one side of the centre wood spine as space permits. Clear nose/main gear and tube roof. Positive restraint in every direction; accessible connector. No pack dimensions or fitted tray claimed.
2 Flight controller	Accessible near the setup balance station on a firm mounting platform, using documented board grommets. Arrow forward; clear USB/SD access, level datum and wires that pull on the board.
3 / 4 / 5 ESC, RX, GPS	ESC aft with airflow and service access; RX on the signal side. Put M9 antenna upward with clear sky; if compass equipped, separate it from battery/current wiring and steel. Place RX active antenna ends clear of carbon and motor power. [16]

Wing pitot placement and harness routes are shown next. Colour marks identify functions; their drawn size and coordinates are not component footprints.

Harness routes and mounting

Keep removable joints accessible; confirm fit before closing skins



Harness	Route and service point
Brown: high current	Battery -> FC distribution -> ESC -> motor. Keep + and - together and away from signal/sensor bundles. Keep ESC input leads short; obtain the actual ESC maker cable-length/capacitor guidance before extending. No unsupported length limit is assumed.
Teal: control	Separate 3-wire S3/S4 wing leads, labelled L / R with signal, servo positive and ground. Retain keyed connectors at wing roots. Route S5/S6 to the tail and S7 to nose steering; secure leads clear of hinges, pushrods and gear.
Blue: navigation	GPS UART and optional compass to FC; airspeed I2C to a supported sensor near its pitot. Mount probe upstream of the wing leading edge in undisturbed flow, away from fuselage boundary layer and prop wash. Keep tubes short, unkinked and removable. Do not choose a fuselage-roof position just because it is convenient. [17]

Install in six passes

1. Trial-fit equipment and mark hatch access.
2. Check battery travel around spine, gear and joiners.
3. Install mounts; separate power and signal wiring.
4. Fit labelled service plugs and strain relief.
5. Check restraint, cooling, linkage travel and antennas with hatches shut.
6. Repeat loaded-power, range, sensor, direction and balance checks.

Trial-fit actual components. The shaded zones are suggested locations, not mounting-hole templates or guaranteed clearances.

[16] ArduPilot GPS/compass mounting

[17] ArduPilot Pitot Tube Considerations

Parts and materials

Quantities are for one complete aircraft. The fit coupon and forming tools are not installed parts.

ID	Part	Qty	Material
A01	Rear joiner support	2	5.000 mm plywood
A04	Aft fuselage former	1	5.000 mm plywood
A05	Front wing spar	2	5.000 mm plywood
A07	Rear wing spar	2	5.000 mm plywood
A10	Outer aileron end plate	2	3.000 mm plywood
A13	Forward longitudinal former	1	5.000 mm plywood
A16	Middle fuselage former	1	5.000 mm plywood
A18	Tail bulkhead	1	2.500 mm plywood
A24	Middle wing rib	2	5.000 mm plywood
A25	Front gear bulkhead	1	5.000 mm plywood
A26	Motor mount	1	5.000 mm plywood
A34	Forward fuselage former	1	5.000 mm plywood
A35	Inner wing rib	2	5.000 mm plywood
A43	Aft longitudinal former	1	5.000 mm plywood
A44	Outer wing rib	2	4.990 mm plywood
A47	Main gear base	1	5.000 mm plywood
A48	Rear gear bulkhead	1	5.000 mm plywood
A53	Inner aileron end plate	2	3.000 mm plywood
A54	Central forward former	1	5.000 mm plywood
A62F	Front nose spine segment	1	5.000 mm plywood
A62R	Rear spine segment	1	5.000 mm plywood
A63	Nose cheek doublers (A63/A64)	2	2.500 mm plywood
A06	Centre fairing	1	LW-PLA preferred; PLA+ mass must be budgeted
A09	Nose steering bearing	1	PLA+ or PETG; functional coupon required

ID	Part	Qty	Material
A11	Left aileron	1	LW-PLA preferred; PLA+ mass must be budgeted
A12	Tail fairing	1	LW-PLA preferred; PLA+ mass must be budgeted
A21	Right aileron	1	LW-PLA preferred; PLA+ mass must be budgeted
A23	Left aileron hinge	1	LW-PLA preferred; PLA+ mass must be budgeted
A36	Right aileron hinge	1	LW-PLA preferred; PLA+ mass must be budgeted
A37	Nose gear mount	1	PLA+ or PETG; functional coupon required
A45	Right wing root	1	PLA+ or PETG; functional coupon required
A55	Left wing root	1	PLA+ or PETG; functional coupon required
A59	Nose shell	1	LW-PLA preferred; PLA+ mass must be budgeted
F01	Left inner wing	1	5 mm FliteBoard Pro
F02	Right inner wing	1	5 mm FliteBoard Pro
F03	Right outer wing	1	5 mm FliteBoard Pro
F04	Left outer wing	1	5 mm FliteBoard Pro
F05	Right fixed tail - main	1	5 mm FliteBoard Pro
F06	Left fixed tail - main	1	5 mm FliteBoard Pro
F07	Right fixed tail - root collar	1	5 mm FliteBoard Pro
F08	Left fixed tail - root collar	1	5 mm FliteBoard Pro
F09	Forward lower fuselage - left curve	1	5 mm FliteBoard Pro
F10	Forward lower fuselage - right curve	1	5 mm FliteBoard Pro
F11	Middle lower fuselage - right curve	1	5 mm FliteBoard Pro
F12	Middle lower fuselage - left curve	1	5 mm FliteBoard Pro
F13	Forward lower fuselage - left lip	1	5 mm FliteBoard Pro
F14	Forward lower fuselage - bottom strip	1	5 mm FliteBoard Pro
F15	Forward lower fuselage - right lip	1	5 mm FliteBoard Pro
F16	Middle lower fuselage - left lip	1	5 mm FliteBoard Pro
F17	Middle lower fuselage - right lip	1	5 mm FliteBoard Pro

ID	Part	Qty	Material
F18	Middle lower fuselage - bottom strip	1	5 mm FliteBoard Pro
F19	Aft lower fuselage - left lip	1	5 mm FliteBoard Pro
F20	Aft lower fuselage - left curve	1	5 mm FliteBoard Pro
F21	Aft lower fuselage - right curve	1	5 mm FliteBoard Pro
F22	Aft lower fuselage - right lip	1	5 mm FliteBoard Pro
F23	Aft upper fuselage - left	1	5 mm FliteBoard Pro
F24	Aft upper fuselage - right	1	5 mm FliteBoard Pro
F25	Forward upper fuselage - right	1	5 mm FliteBoard Pro
F26	Forward upper fuselage - left	1	5 mm FliteBoard Pro
F27	Rear underside band 1 - forward	1	5 mm FliteBoard Pro
F28	Rear underside band 2	1	5 mm FliteBoard Pro
F29	Rear underside band 3	1	5 mm FliteBoard Pro
F30	Rear underside band 4 - aft	1	5 mm FliteBoard Pro
F31	Right tail control laminate blank	2	5 mm FliteBoard Pro
F32	Left tail control laminate blank	2	5 mm FliteBoard Pro
F33	Aft lower fuselage - bottom strip	1	5 mm FliteBoard Pro
T01	T01 front joiner	1	Carbon OD8/ID6 mm
T02	T02 right front sleeve	1	Carbon OD10/ID8.2 mm
T03	T03 left front sleeve	1	Carbon OD10/ID8.2 mm
T04	T04 rear joiner	1	Carbon OD8/ID6 mm
H01	Main formed landing gear	1	12.7 x3.175 mm suitable formed metal
H02	Nose wire extended	1	3 mm spring steel wire
H03	Wheels	3	50.8 mm diameter,3 mm bore
H04	Servos	5	9 g metal gear; actual model fit/voltage check
H05	Control horns	4	Purchased lightweight RC horns
H06	Nose steering arm	1	3 mm clamping arm

ID	Part	Qty	Material
H07	Pushrod and clevis sets	5	Matched RC linkage
H08	Aileron pins	2	3 mm smooth rod; start 285 mm
H09	Fastener pack	1	See airframe guide page 3 and hardware instructions

Hardware and forming

A02 main landing gear

formed strip reference, not a plastic print and not a laser flat blank. Source section is 12.7 mm fore-aft width by 3.175 mm strip thickness. Overall stance approx 285.5 mm; main axle centres $x=\pm 151$ mm with wheels, axle height $z=-135.85$. Use a suitable formed model-aircraft metal gear or commission this geometry. The metalwork shop must select a bendable alloy/temper and finite bend radii appropriate to it, then preserve attachment and axle stations. Source CAD has ideal sharp bends; do not interpret these as zero-radius physical bends. Start with a strip longer than developed centreline, form to the STEP/template and finish ends. Shop calculates bend deductions for its tooling. Two new 3.2 mm attachment holes $x=\pm 22.8222, y=136.35$ align A47; fit 2xM3 through-bolts, large washers and locking nuts. Suitable small-aircraft landing gear requires physical load/deflection testing for completed aircraft mass.

A46 nose wire

3 mm spring/piano-wire reference, with upper shaft extended to $z=+1$ mm so it reaches the steering bearing/arm. Preserve the source bends and axle line. Cut to the supplied formed STEP using a bending jig; final upper end can be trimmed only after bearing, shaft collar and steering arm are installed. Wire must rotate freely in finished 3.2 mm bearing bores without lateral play that permits wheel shimmy. Do not 3D print it. Final upper length depends on actual clamp/arm.

Wheels

3 purchased lightweight 50.8 mm diameter, approximately 16.5 mm wide, 3 mm bore. Two main axle bolts nominal M3x 25 mm plus spacers/washers/locking nuts; nose wheel uses wire axle with 3 mm collars. Fit freely rotating wheels without clamping hubs.

Controls

2x 3 mm smooth hinge rods, start 285 mm long, final trim flush to outer end retainers with approximately 1 mm axial freedom. 4 purchased lightweight horns for 2 ailerons+2 ruddervators; 1 clamping 3 mm nose-steering arm. 5 pushrods with matching clevises/ball links, cut only after neutral servo and surface positions are set. Do not use glue alone to retain hinge pins or wing joiners.

Root wing retention

six M2 through-bolts, start 20 mm length, washers and locking nuts in three aligned root attachment holes per wing. Finish printed/wood holes to the actual fastener clearance. Set lengths to avoid moving linkages. These root fasteners retain removable wings; do not glue the sliding carbon joiner into both wings.

Motor

four M3 bolts through revised A26, 19 mm PCD. Select length to pass 5 mm plywood+washer and engage motor threads without reaching windings. Actual screw engagement depth must be measured; do not blindly reuse a long screw. Manufacturer drawing is supplied.

The 80x 35x 4 mm print coupon has bores 2.0, 2.2, 3.0, 3.2, 8.0, 8.2, 10.2 mm. Use it to establish printer/material fit; it is not an aircraft part. Size must remain 100%. Verify outside dimensions and fit

using intended rods/bolts before printing structural brackets.

Nose bearing A09 to mount A37

four M3 machine screws with washers and locking nuts, start 16-20 mm long and select after measuring the actual stack. The shaft-through portion of the mount is 7 mm and the bearing foot about 3 mm, plus washer/nut stack. Use the printed underside access recesses; do not trap a loose nut where it cannot be inspected. Finish the bearing clearance holes to the actual screw and verify no bolt can touch the rotating nose shaft.

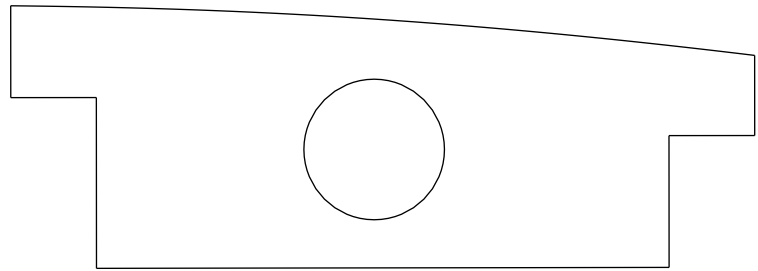
MQ9 manufacturing profiles - 22 profiles / 31 pieces

Preview only; DXF units mm, 1:1 native curves. Qty includes inferred mirrored wing members. Declared bore revisions and prismatic blanks; CUT layer only.

A01 MQ9 Assembled v28

5.000 mm | 43.40 x 15.32 mm | Qty 2 | 1 internal loops

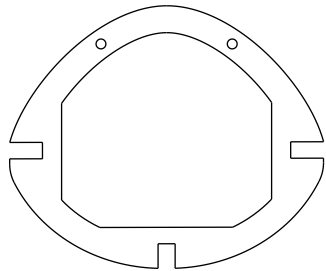
REVISED MANUFACTURING PROFILE



A04 Aft_Fuselage_Hoop v1

5.000 mm | 95.65 x 80.03 mm | Qty 1 | 3 internal loops

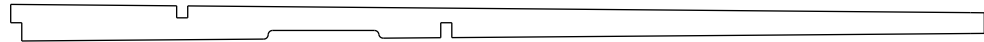
REVISED MANUFACTURING PROFILE



A05 Front_Spar v1

5.000 mm | 440.00 x 16.54 mm | Qty 2 | 0 internal loops

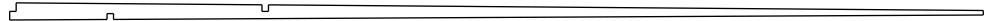
SOURCE CONTOUR



A07 Rear_Spar v1

5.000 mm | 750.00 x 13.19 mm | Qty 2 | 0 internal loops

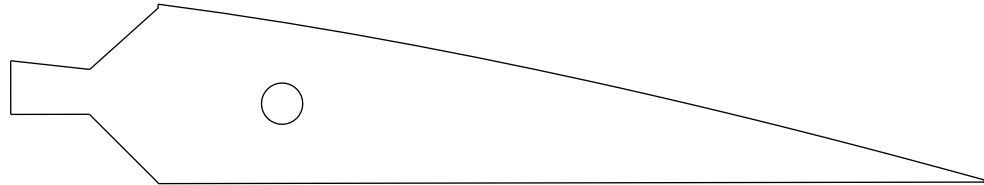
SOURCE CONTOUR



A10 Outer_Aileron_Plate v1

3.000 mm | 70.72 x 13.03 mm | Qty 2 | 1 internal loops

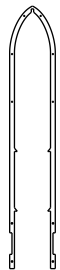
PRISMATIC BLANK - FINISH EDGE



A13 Front_Lateral_Hoop v1

5.000 mm | 95.64 x 550.66 mm | Qty 1 | 10 internal loops

SOURCE CONTOUR



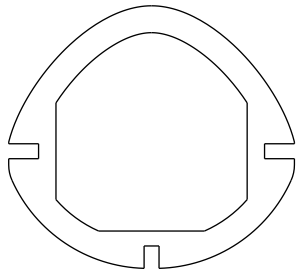
MQ9 manufacturing profiles - 22 profiles / 31 pieces

Preview only; DXF units mm, 1:1 native curves. Qty includes inferred mirrored wing members. Declared bore revisions and prismatic blanks; CUT layer only.

A16 Fwd_Fuselage_Hoop v1 (2)

5.000 mm | 95.64 x 87.90 mm | Qty 1 | 1 internal loops

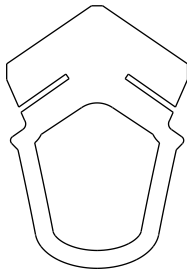
SOURCE CONTOUR



A18 MQ9 Assembled v28

2.500 mm | 81.00 x 117.59 mm | Qty 1 | 1 internal loops

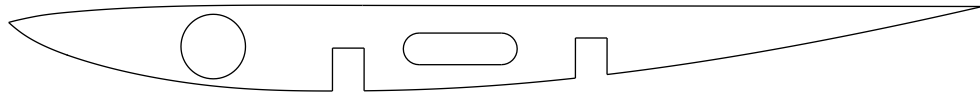
SOURCE CONTOUR



A24 Middle_Rib v1

5.000 mm | 153.89 x 13.56 mm | Qty 2 | 2 internal loops

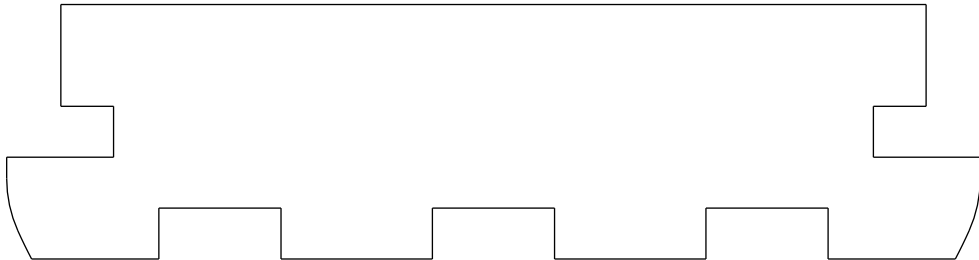
REVISED MANUFACTURING PROFILE



A25 MQ9 Assembled v28

5.000 mm | 95.64 x 25.00 mm | Qty 1 | 0 internal loops

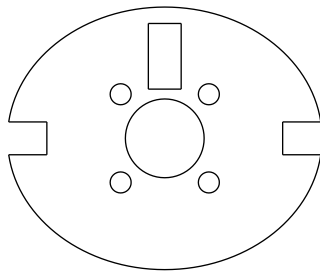
SOURCE CONTOUR



A26 Notor_Mount v1

5.000 mm | 47.55 x 40.00 mm | Qty 1 | 6 internal loops

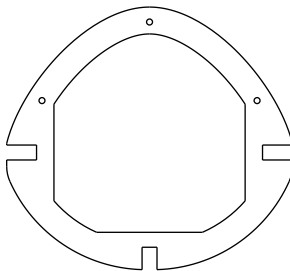
REVISED MANUFACTURING PROFILE



A34 Fwd_Fuselage_Hoop v1 (1)

5.000 mm | 95.64 x 87.90 mm | Qty 1 | 4 internal loops

SOURCE CONTOUR



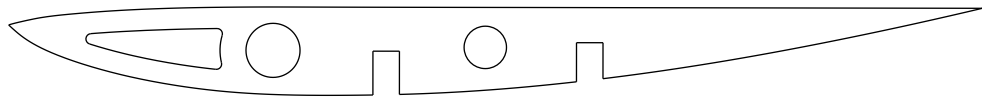
MQ9 manufacturing profiles - 22 profiles / 31 pieces

Preview only; DXF units mm, 1:1 native curves. Qty includes inferred mirrored wing members. Declared bore revisions and prismatic blanks; CUT layer only.

A35 Inner_Rib v1

5.000 mm | 183.66 x 16.67 mm | Qty 2 | 3 internal loops

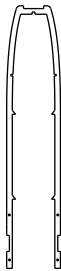
REVISED MANUFACTURING PROFILE



A43 Aft_Lateral_Hoop v1

5.000 mm | 95.64 x 467.00 mm | Qty 1 | 4 internal loops

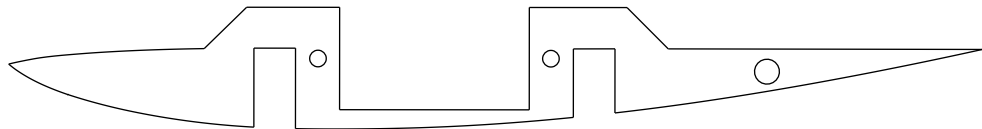
SOURCE CONTOUR



A44 Outer_Rib v1

4.990 mm | 117.03 x 14.63 mm | Qty 2 | 3 internal loops

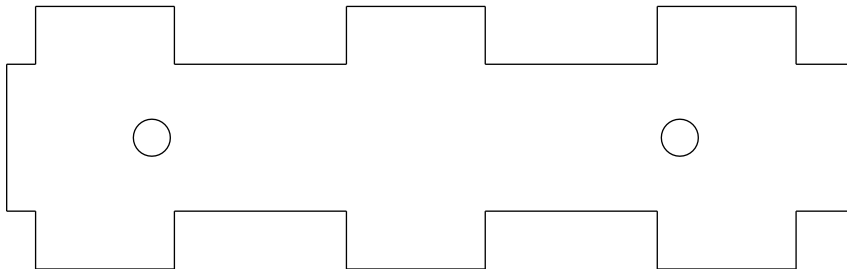
PRISMATIC BLANK - FINISH EDGE



A47 MQ9 Assembled v28

5.000 mm | 72.74 x 22.70 mm | Qty 1 | 2 internal loops

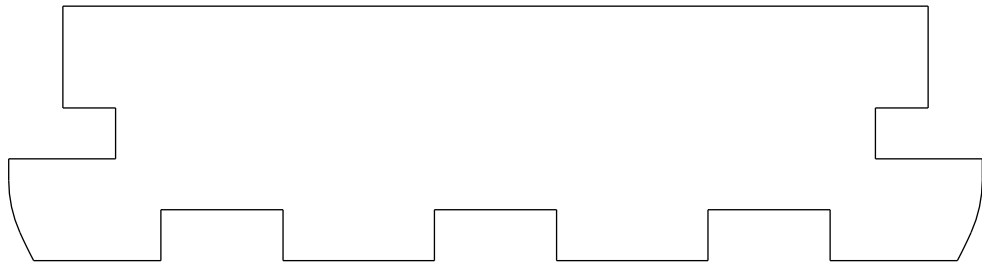
REVISED MANUFACTURING PROFILE



A48 MQ9 Assembled v28

5.000 mm | 95.64 x 25.00 mm | Qty 1 | 0 internal loops

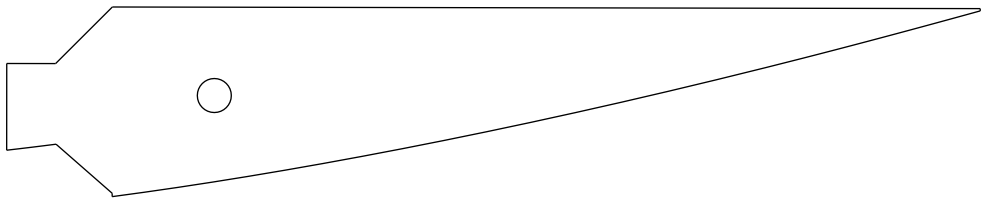
SOURCE CONTOUR



A53 Inner_Aileron_Plate v1

3.000 mm | 85.87 x 16.73 mm | Qty 2 | 1 internal loops

SOURCE CONTOUR



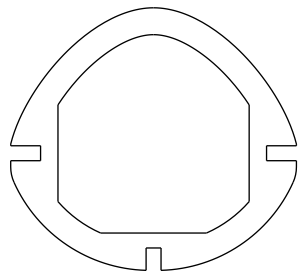
MQ9 manufacturing profiles - 22 profiles / 31 pieces

Preview only; DXF units mm, 1:1 native curves. Qty includes inferred mirrored wing members. Declared bore revisions and prismatic blanks; CUT layer only.

A54 Fwd_Fuselage_Hoop v1

5.000 mm | 95.64 x 87.90 mm | Qty 1 | 1 internal loops

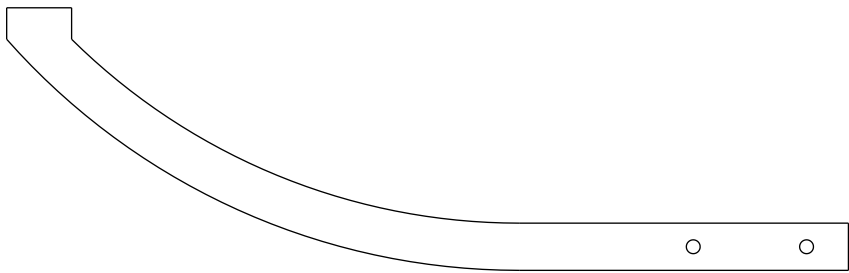
SOURCE CONTOUR



A62F Unified spine forward segment

5.000 mm | 133.72 x 41.71 mm | Qty 1 | 2 internal loops

REVISED MANUFACTURING PROFILE



A62R Unified spine rear segment

5.000 mm | 887.62 x 41.71 mm | Qty 1 | 2 internal loops

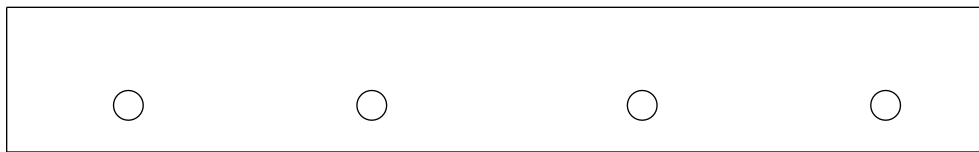
REVISED MANUFACTURING PROFILE



A63 Nose gear continuous side doubler pair (A63/A64)

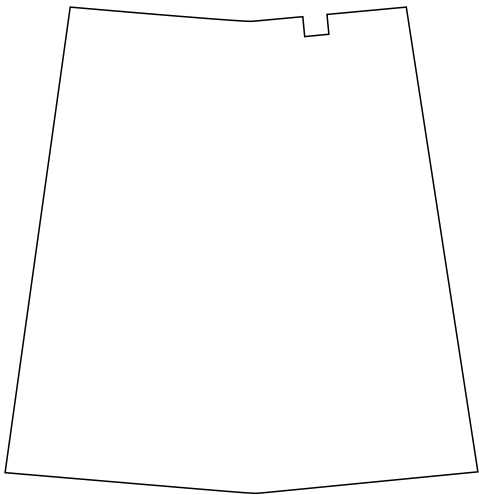
2.500 mm | 72.00 x 10.70 mm | Qty 2 | 4 internal loops

REVISED MANUFACTURING PROFILE

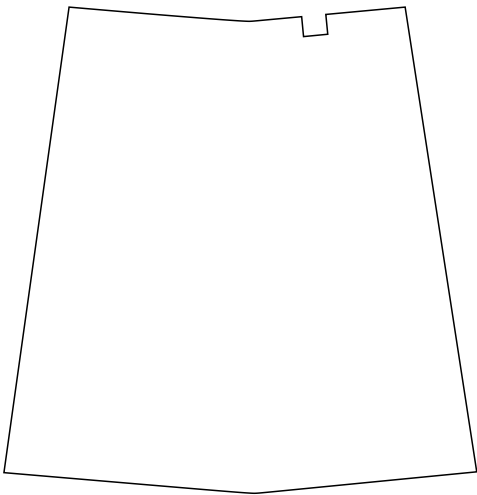


MQ9 | developed foam patterns | overview only

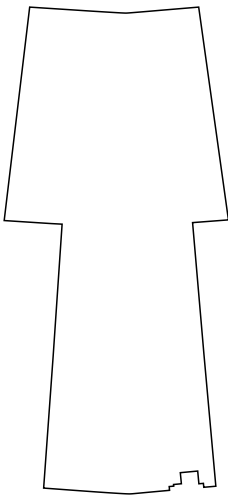
Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



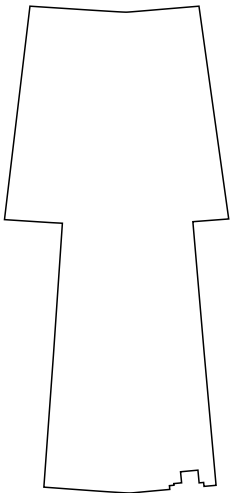
F01 Developed skin
wing_inner_A
443.1 x 455.6 mm | max strain 0.341%



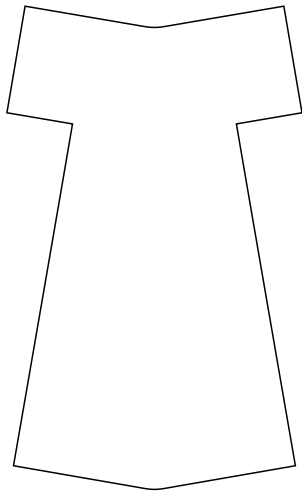
F02 Developed skin
wing_inner_B
443.1 x 455.6 mm | max strain 0.341%



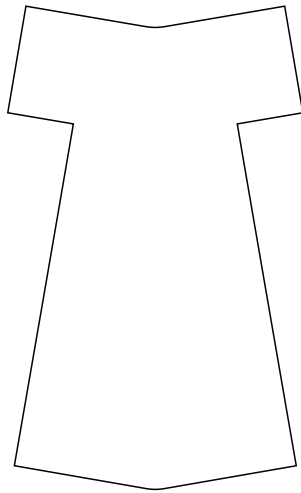
F03 Developed skin
wing_outer_A
237.1 x 514.7 mm | max strain 0.357%



F04 Developed skin
wing_outer_B
237.1 x 514.7 mm | max strain 0.366%



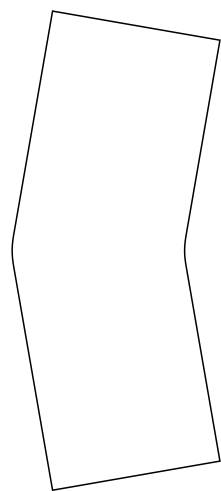
F05 Developed skin
tail_A_rootcollar1
177.4 x 290.5 mm | max strain 0.278%



F06 Developed skin
tail_B_rootcollar1
177.4 x 290.5 mm | max strain 0.278%

MQ9 | developed foam patterns | overview only

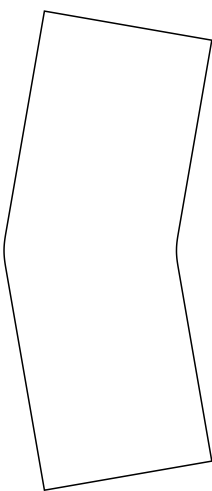
Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



F07 Developed skin

tail_A_rootcollar2

83.6 x 192.9 mm | max strain 0.174%



F08 Developed skin

tail_B_rootcollar2

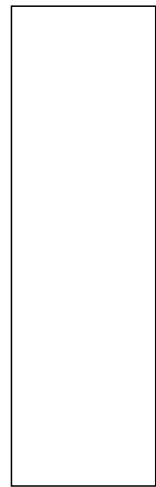
83.6 x 192.9 mm | max strain 0.174%



F09 Developed skin

fuselage_forward_A

61.6 x 205.0 mm | max strain 0.023%



F10 Developed skin

fuselage_forward_B

61.6 x 205.0 mm | max strain 0.023%



F11 Developed skin

fuselage_middle_A

61.6 x 368.7 mm | max strain 0.023%



F12 Developed skin

fuselage_middle_B

61.6 x 368.6 mm | max strain 0.023%

MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



F13 Developed skin

fuselage_src3_face8
14.4 x 205.0 mm | max strain 0.018%



F14 Developed skin

fuselage_src3_face10
5.3 x 205.0 mm | max strain 0.000%



F15 Developed skin

fuselage_src3_face12
14.4 x 205.0 mm | max strain 0.019%



F16 Developed skin

fuselage_src21_face2
14.4 x 368.7 mm | max strain 0.018%



F17 Developed skin

fuselage_src21_face16
14.4 x 368.7 mm | max strain 0.019%

F18 Developed skin

fuselage_src21_face18
5.3 x 368.7 mm | max strain 0.000%

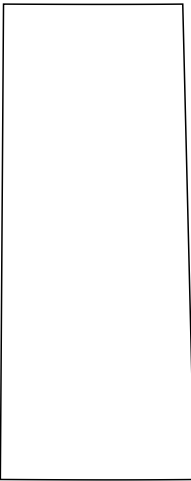
MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



F19 Developed skin

fuselage_src10_face16
14.6 x 151.0 mm | max strain 0.000%



F20 Developed skin

fuselage_src10_face17
61.6 x 151.1 mm | max strain 0.037%



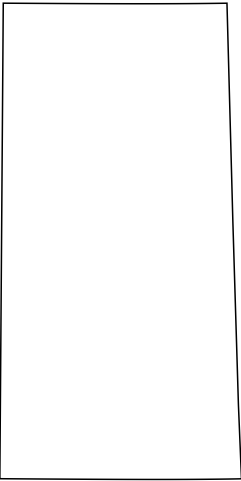
F21 Developed skin

fuselage_src10_face19
61.8 x 151.4 mm | max strain 0.035%



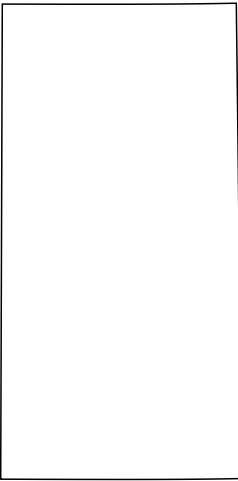
F22 Developed skin

fuselage_src10_face20
14.8 x 151.0 mm | max strain 0.000%



F23 Developed skin

fuselage_src22_face11
79.3 x 156.2 mm | max strain 0.030%

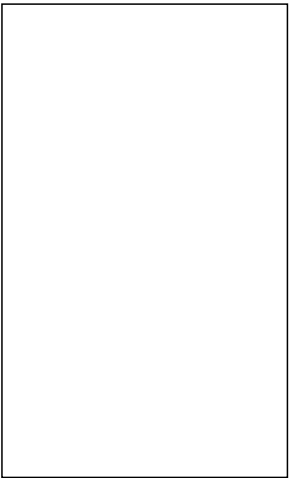


F24 Developed skin

fuselage_src22_face13
78.8 x 156.4 mm | max strain 0.037%

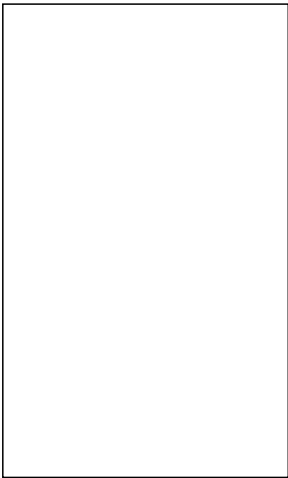
MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



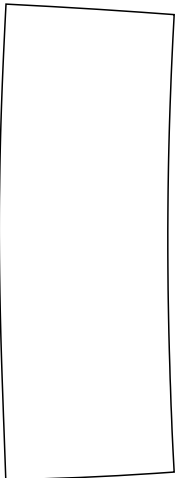
F25 Developed skin

fuselage_src25_face6
78.4 x 130.0 mm | max strain 0.000%



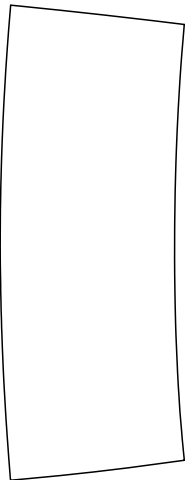
F26 Developed skin

fuselage_src25_face8
78.4 x 130.0 mm | max strain 0.000%



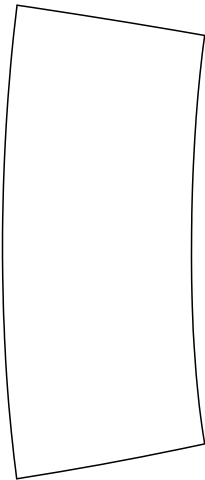
F27 Developed skin

fuselage_src18_face14_axis1_strip1of4
55.2 x 150.2 mm | max strain 0.287%



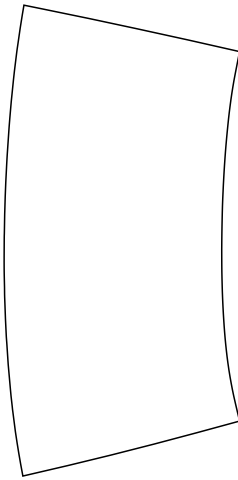
F28 Developed skin

fuselage_src18_face14_axis1_strip2of4
55.9 x 144.4 mm | max strain 0.295%



F29 Developed skin

fuselage_src18_face14_axis1_strip3of4
56.5 x 132.3 mm | max strain 0.336%



F30 Developed skin

fuselage_src18_face14_axis1_strip4of4
57.0 x 113.9 mm | max strain 0.321%

MQ9 | developed foam patterns | overview only

Not to scale. 5mm board; preserved outer paper forming method. All numerical edge strains below 0.5%.



F31 Control blank

tail_control_blank_A
45.0 x 275.0 mm | max strain 0.000%



F32 Control blank

tail_control_blank_B
45.0 x 275.0 mm | max strain 0.000%

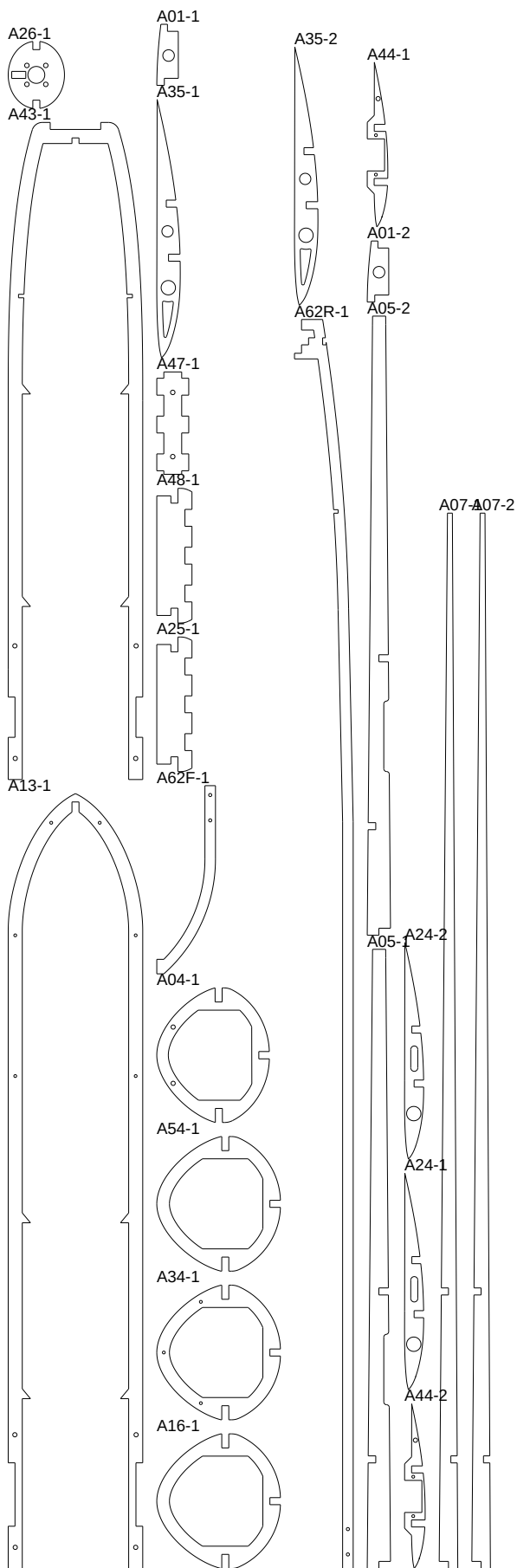


F33 Developed skin

fuselage_src10_face18
5.3 x 151.1 mm | max strain 0.000%

5mm stock: 24 pieces

Reference sheet1220x2440mm. Compact work area 362.8x1117.4mm. Grain +Y.



3mm stock: 4 pieces

Reference sheet1220x2440mm. Compact work area 36.7x363.2mm. Grain +Y.

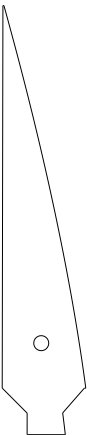
A10-2



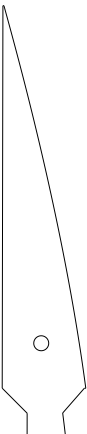
A10-1



A53-2



A53-1



2.5mm stock: 3 pieces

Reference sheet1220x2440mm. Compact work area 101.0x301.6mm. Grain +Y.

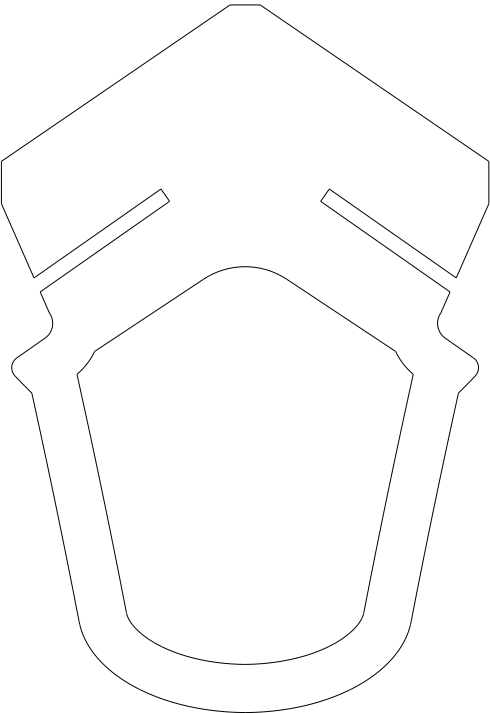
A63-2



A63-1

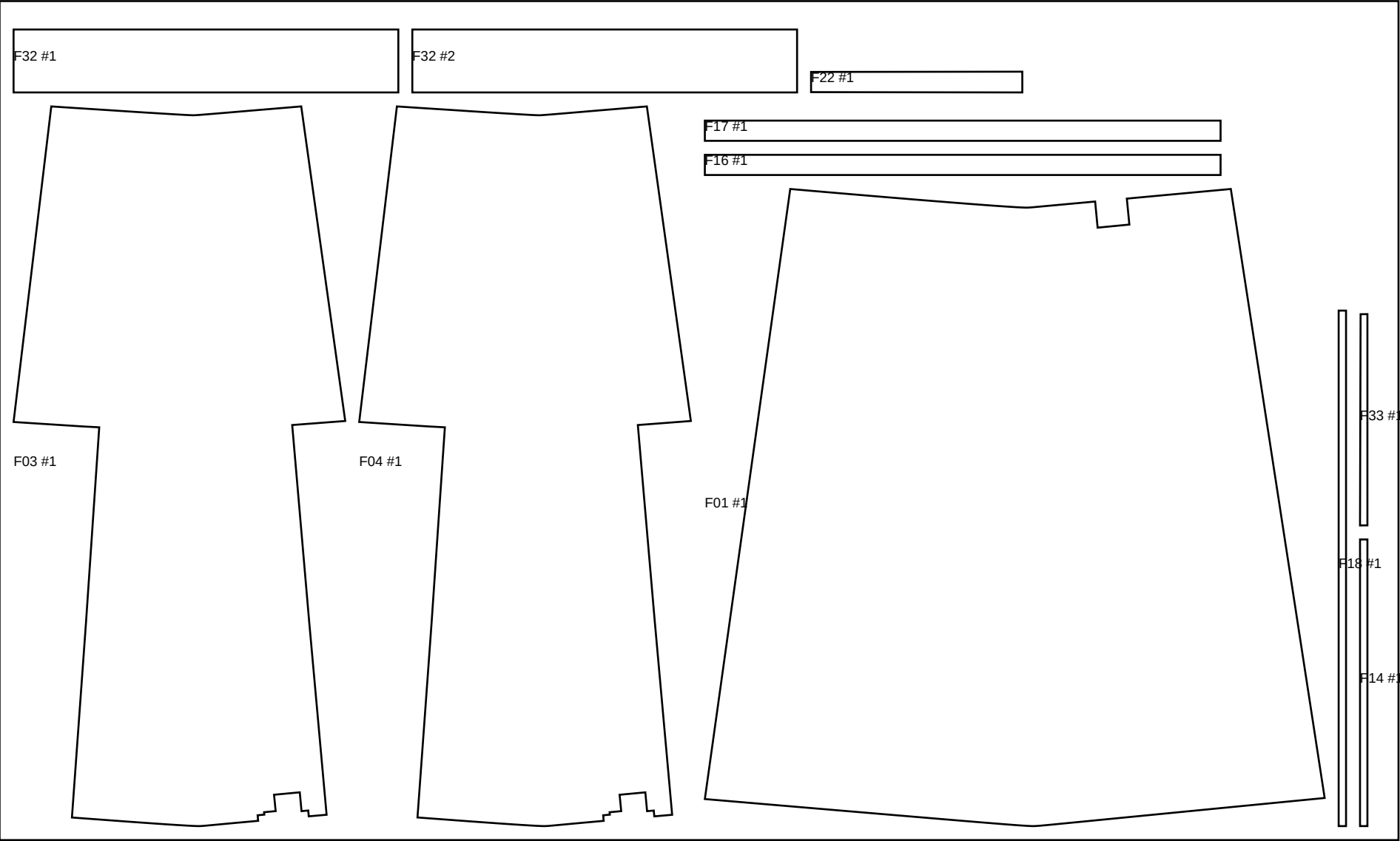


A18-1



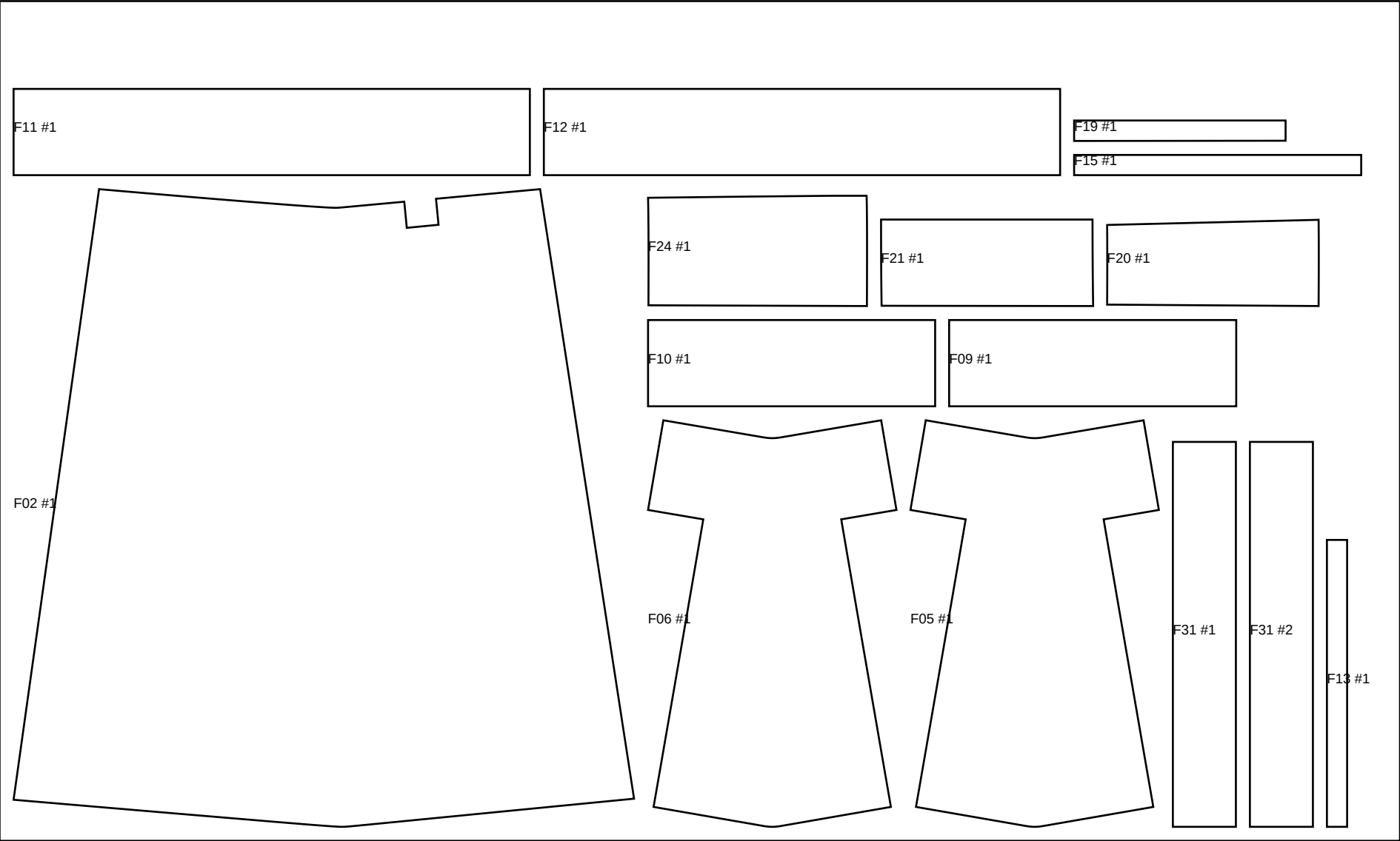
Foam sheet 1 / 1000 x 600 x 5 mm

Overview only. Knife cut at 100% from DXF or full-size pattern tiles. CUT layer only.



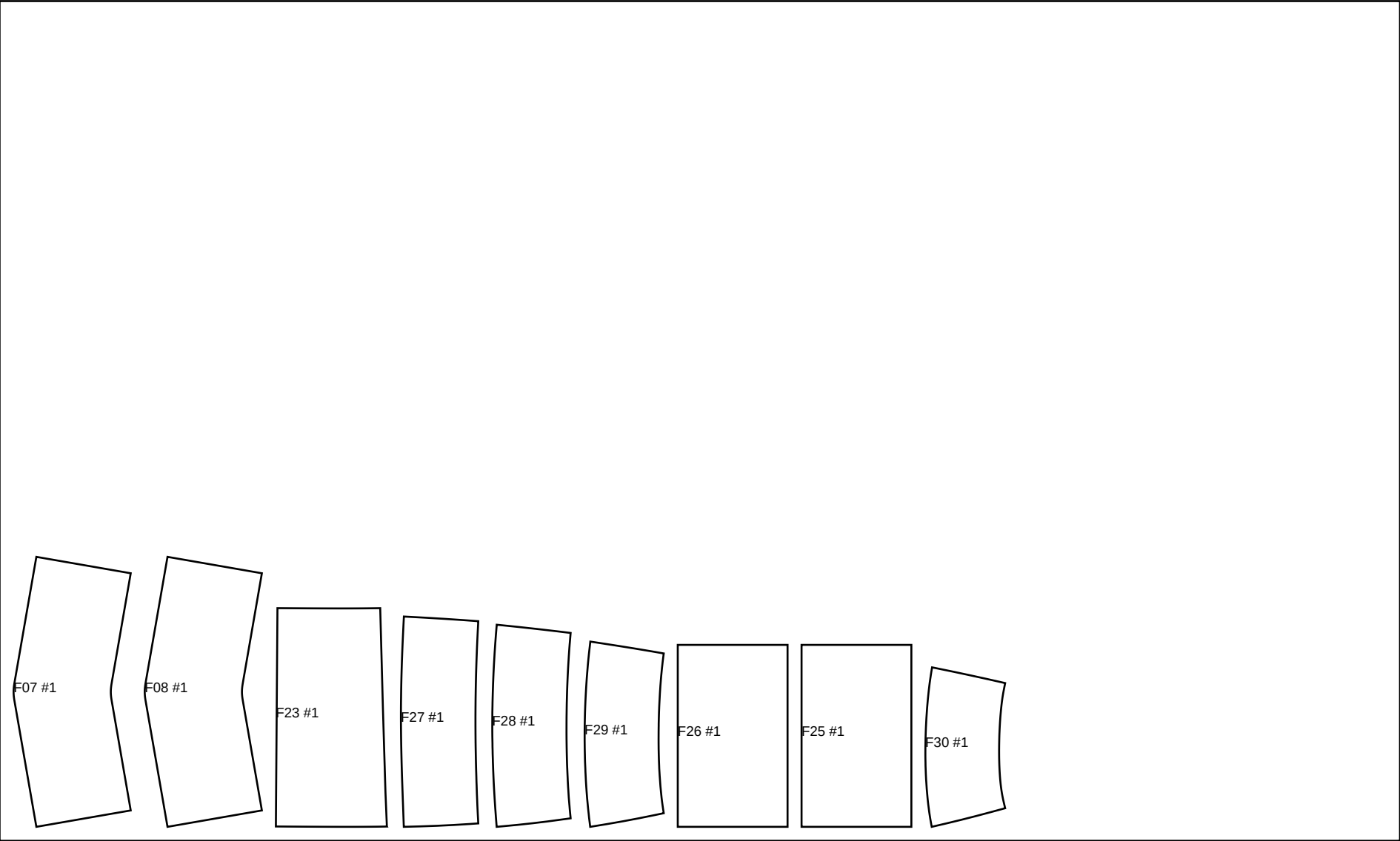
Foam sheet 2 / 1000 x 600 x 5 mm

Overview only. Knife cut at 100% from DXF or full-size pattern tiles. CUT layer only.



Foam sheet 3 / 1000 x 600 x 5 mm

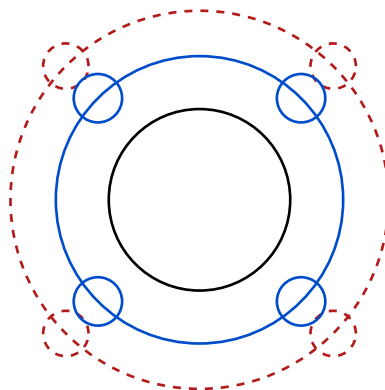
Overview only. Knife cut at 100% from DXF or full-size pattern tiles. CUT layer only.



A26 motor mount - declared drilling correction

Source: 4x3mm on 25mm PCD. Revised: 4x3.2mm on 19mm PCD.

Physical Emax motor check required before drilling; drawing below at 2:1.



Red dashed: obsolete 25mm PCD - DO NOT DRILL

Blue: 19mm PCD - four 3.2mm holes

Central 12mm hole unchanged. Minimum radial wood web 1.9mm; use washers.

Checks and build acceptance

Read this before ordering full production or planning a flight.

Recorded digital checks

Check	Result
Plywood	22 profiles / 31 pieces; mirrored quantities and revised joints checked.
DXF	66 delivered DXFs checked for millimetre units and file validity.
Printed parts	11 aircraft meshes plus coupon: closed, connected and manifold. Maximum mesh/CAD volume difference 0.052%.
Foam development	33 patterns / 35 blanks. Maximum sampled edge strain 0.367%; seam mismatch 0.027 mm. Physical forming is still required.
Paper templates	Scale, registration and part placement checked. 7 plywood pages, 23 foam pages, 2 tooling pages.

Manufacturing Team

Measure stock; run slot, printer and foam coupons. Dry-fit all parts together. Check bonded joints, wing retention, gear deflection and moving clearances. Record completed masses and any fitting work.

Electronics Team

Identify exact connectors and servo ratings. Verify loaded supply voltage, controls, sensors, range and failsafes. Measure propulsion current, thrust and temperature. Check final mass and balance.

Before flight

The 522 mm aft-nose mark is a provisional geometric reference, not an approved flight CG range. The 9-inch propeller / 40 A ESC combination remains unqualified. Have an experienced fixed-wing pilot review the completed aircraft and its test results.